EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #398 *Topic 1*

A company needs to transfer 600 TB of data from its on-premises network-attached storage (NAS) system to the AWS Cloud. The data transfer must be complete within 2 weeks. The data is sensitive and must be encrypted in transit. The company's internet connection can support an upload speed of 100 Mbps.

Which solution meets these requirements MOST cost-effectively?

A. Use Amazon S3 multi-part upload functionality to transfer the files over HTTPS.

B. Create a VPN connection between the on-premises NAS system and the nearest AWS Region. Transfer the data over the VPN connection.

C. Use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices. Use the devices to transfer the data to Amazon S3. **Most Voted**

D. Set up a 10 Gbps AWS Direct Connect connection between the company location and the nearest AWS Region. Transfer the data over a VPN connection into the Region to store the data in Amazon S3.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

shanwford Highly Voted 1 year, 1 month ago

Selected Answer: C

With the existing data link the transfer takes ~ 600 days in the best case. Thus, (A) and (B) are not applicable. Solution (D) could meet the target with a transfer time of 6 days, but the lead time for the direct connect deployment can take weeks! Thus, (C) is the only valid solution.

upvoted 14 times

Guru4Cloud Most Recent 9 months, 1 week ago

Selected Answer: C

Use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices. Use the devices to transfer the data to Amazon S3.

upvoted 2 times

TariqKipkemei 1 year ago

Selected Answer: C

C is the best option considering the time and bandwidth limitations
upvoted 2 times

pbpally 1 year, 1 month ago

Selected Answer: C

We need the admin in here to tell us how they plan on this being achieved over connection with such a slow connection lol.
It's C, folks.
upvoted 3 times

KAUS2 1 year, 2 months ago

Selected Answer: C

Best option is to use multiple AWS Snowball Edge Storage Optimized devices. Option "C" is the correct one.
upvoted 2 times

ktulu2602 1 year, 2 months ago

Selected Answer: C

All others are limited by the bandwidth limit
upvoted 1 times

ktulu2602 1 year, 2 months ago

Or provisioning time in the D case
upvoted 2 times

KZM 1 year, 3 months ago

It is C. Snowball (from Snow Family).
upvoted 2 times

cegama543 1 year, 3 months ago

Selected Answer: C

C. Use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices. Use the devices to transfer the data to Amazon S3.

The best option is to use the AWS Snow Family console to order several AWS Snowball Edge Storage Optimized devices and use the devices to transfer the data to Amazon S3. Snowball Edge is a petabyte-scale data transfer device that can help transfer large amounts of data securely and quickly. Using Snowball Edge can be the most cost-effective solution for transferring large amounts of data over long distances and can help meet the requirement of transferring 600 TB of data within two weeks.
upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #399 *Topic 1*

A financial company hosts a web application on AWS. The application uses an Amazon API Gateway Regional API endpoint to give users the ability to retrieve current stock prices. The company's security team has noticed an increase in the number of API requests. The security team is concerned that HTTP flood attacks might take the application offline.

A solutions architect must design a solution to protect the application from this type of attack.

Which solution meets these requirements with the LEAST operational overhead?

A. Create an Amazon CloudFront distribution in front of the API Gateway Regional API endpoint with a maximum TTL of 24 hours.

B. Create a Regional AWS WAF web ACL with a rate-based rule. Associate the web ACL with the API Gateway stage.
Most Voted

C. Use Amazon CloudWatch metrics to monitor the Count metric and alert the security team when the predefined rate is reached.

D. Create an Amazon CloudFront distribution with Lambda@Edge in front of the API Gateway Regional API endpoint. Create an AWS Lambda function to block requests from IP addresses that exceed the predefined rate.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

Guru4Cloud Highly Voted 9 months, 1 week ago

Selected Answer: B

Regional AWS WAF web ACL is a managed web application firewall that can be used to protect your API Gateway API from a variety of attacks, including HTTP flood attacks.
Rate-based rule is a type of rule that can be used to limit the number of requests that can be made from a single IP address within a specified period of time.
API Gateway stage is a logical grouping of API resources that can be used to control access to your API.

upvoted 10 times

elearningtakai Highly Voted 1 year, 2 months ago

Selected Answer: B

Selected Answer: B

A rate-based rule in AWS WAF allows the security team to configure thresholds that trigger rate-based rules, which enable AWS WAF to track the rate of requests for a specified time period and then block them automatically when the threshold is exceeded. This provides the ability to prevent HTTP flood attacks with minimal operational overhead.
upvoted 6 times

TariqKipkemei Most Recent 1 year ago

Selected Answer: B

Answer is B
upvoted 2 times

maxicalypse 1 year, 2 months ago

B os correct
upvoted 2 times

kampatra 1 year, 2 months ago

Selected Answer: B

https://docs.aws.amazon.com/waf/latest/developerguide/web-acl.html
upvoted 2 times

[Removed] 1 year, 3 months ago

Selected Answer: B

bbbbbbbb
upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #400 *Topic 1*

A meteorological startup company has a custom web application to sell weather data to its users online. The company uses Amazon DynamoDB to store its data and wants to build a new service that sends an alert to the managers of four internal teams every time a new weather event is recorded. The company does not want this new service to affect the performance of the current application.

What should a solutions architect do to meet these requirements with the LEAST amount of operational overhead?

A. Use DynamoDB transactions to write new event data to the table. Configure the transactions to notify internal teams.

B. Have the current application publish a message to four Amazon Simple Notification Service (Amazon SNS) topics. Have each team subscribe to one topic.

C. Enable Amazon DynamoDB Streams on the table. Use triggers to write to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe. **Most Voted**

D. Add a custom attribute to each record to flag new items. Write a cron job that scans the table every minute for items that are new and notifies an Amazon Simple Queue Service (Amazon SQS) queue to which the teams can subscribe.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 1 year, 2 months ago

Selected Answer: C

The best solution to meet these requirements with the least amount of operational overhead is to enable Amazon DynamoDB Streams on the table and use triggers to write to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe. This solution requires minimal configuration and infrastructure setup, and Amazon DynamoDB Streams provide a low-latency way to capture changes to the DynamoDB table. The triggers automatically capture the changes and publish them to the SNS topic, which notifies the internal teams.

upvoted 15 times

Buruguduystunstugudunstuy 1 year, 2 months ago

Answer A is not a suitable solution because it requires additional configuration to notify the internal teams, and it could add operational overhead to the application.

Answer B is not the best solution because it requires changes to the current application, which may affect its performance, and it creates additional work for the teams to subscribe to multiple topics.

Answer D is not a good solution because it requires a cron job to scan the table every minute, which adds additional operational overhead to the system.

Therefore, the correct answer is C. Enable Amazon DynamoDB Streams on the table. Use triggers to write to a single Amazon SNS topic to which the teams can subscribe.

upvoted 6 times

Manimgh Most Recent 1 month, 3 weeks ago

Selected Answer: C

Sends an alert = SNS

upvoted 1 times

Guru4Cloud 9 months, 1 week ago

Selected Answer: C

Enable Amazon DynamoDB Streams on the table. Use triggers to write to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe

upvoted 4 times

james2033 10 months, 3 weeks ago

Selected Answer: C

Question keyword: "sends an alert", a new weather event is recorded". Answer keyword C "Amazon DynamoDB Streams on the table", "Amazon Simple Notification Service" (Amazon SNS). Choose C. Easy question.

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Streams.html

https://aws.amazon.com/blogs/database/dynamodb-streams-use-cases-and-design-patterns/

upvoted 4 times

TariqKipkemei 1 year ago

Selected Answer: C

Best answer is C

upvoted 3 times

TariqKipkemei 7 months, 2 weeks ago

DynamoDB Streams captures a time-ordered sequence of item-level modifications in any DynamoDB table and stores this information in a log for up to 24 hours. This capture activity can also invoke triggers to write the event to a single Amazon Simple Notification Service (Amazon SNS) topic to which the teams can subscribe to.

upvoted 6 times

Hemanthgowda1932 1 year, 2 months ago

C is correct

upvoted 2 times

Santosh43 1 year, 2 months ago

definitely C

upvoted 2 times

Bezha 1 year, 2 months ago

Selected Answer: C

DynamoDB Streams

upvoted 4 times

sitha 1 year, 2 months ago

Selected Answer: C

Answer : C

upvoted 2 times

upvoted 2 times

[Removed] 1 year, 3 months ago

Selected Answer: C

cccccccc

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #401 *Topic 1*

A company wants to use the AWS Cloud to make an existing application highly available and resilient. The current version of the application resides in the company's data center. The application recently experienced data loss after a database server crashed because of an unexpected power outage.

The company needs a solution that avoids any single points of failure. The solution must give the application the ability to scale to meet user demand.

Which solution will meet these requirements?

A. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration. **Most Voted**

B. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group in a single Availability Zone. Deploy the database on an EC2 instance. Enable EC2 Auto Recovery.

C. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance with a read replica in a single Availability Zone. Promote the read replica to replace the primary DB instance if the primary DB instance fails.

D. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Deploy the primary and secondary database servers on EC2 instances across multiple Availability Zones. Use Amazon Elastic Block Store (Amazon EBS) Multi-Attach to create shared storage between the instances.

Correct Answer: *A*

Community vote distribution

A (93%) Other (7%)

Comments

pentium75 Highly Voted 11 months, 1 week ago

Selected Answer: A

B has app servers in a single AZ and a database on a single instance
C has both DB replicas in a single AZ
D does not work (EBS Multi-Attach requires EC2 instances in same AZ), and if it would work then the EBS volume would be an SPOF

upvoted 7 times

Guru4Cloud Most Recent 1 year, 3 months ago

Selected Answer: A

Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration

upvoted 2 times

czyboi 1 year, 4 months ago

Why is C incorrect ?

upvoted 1 times

Guru4Cloud 1 year, 3 months ago

C is incorrect because the read replica also resides in a single AZ

upvoted 4 times

antropaws 1 year, 6 months ago

Selected Answer: A

A most def.

upvoted 3 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: A

Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration.

upvoted 3 times

Buruguduystunstugudunstuy 1 year, 8 months ago

Selected Answer: A

The correct answer is A. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration.

To make an existing application highly available and resilient while avoiding any single points of failure and giving the application the ability to scale to meet user demand, the best solution would be to deploy the application servers using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones and use an Amazon RDS DB instance in a Multi-AZ configuration.

By using an Amazon RDS DB instance in a Multi-AZ configuration, the database is automatically replicated across multiple Availability Zones, ensuring that the database is highly available and can withstand the failure of a single Availability Zone. This provides fault tolerance and avoids any single points of failure.

upvoted 3 times

Thief 1 year, 8 months ago

Selected Answer: D

Why not D?

upvoted 1 times

Guru4Cloud 1 year, 3 months ago

D is incorrect because using Multi-Attach EBS adds complexity and doesn't provide automatic DB failover

upvoted 2 times

pentium75 11 months, 1 week ago

Multi-Attach does not work across Availability Zones.

upvoted 2 times

Buruguduystunstugudunstuy 1 year, 8 months ago

Answer D, deploying the primary and secondary database servers on EC2 instances across multiple Availability Zones and using Amazon Elastic Block Store (Amazon EBS) Multi-Attach to create shared storage between the instances, may provide high availability for the database but may introduce additional complexity, and management overhead, and potential

performance issues.
upvoted 2 times

WherecanIstart 1 year, 8 months ago

Selected Answer: A

Highly available = Multi-AZ approach
upvoted 3 times

nileshlg 1 year, 8 months ago

Selected Answer: A

Answers is A
upvoted 1 times

[Removed] 1 year, 8 months ago

Selected Answer: A

Option A is the correct solution. Deploying the application servers in an Auto Scaling group across multiple Availability Zones (AZs) ensures high availability and fault tolerance. An Auto Scaling group allows the application to scale horizontally to meet user demand. Using Amazon RDS DB instance in a Multi-AZ configuration ensures that the database is automatically replicated to a standby instance in a different AZ. This provides database redundancy and avoids any single point of failure.
upvoted 2 times

quentin17 1 year, 8 months ago

Selected Answer: C

Highly available
upvoted 1 times

pentium75 11 months, 1 week ago

No because instance and read replica "in a single Availability Zone"
upvoted 2 times

KAUS2 1 year, 9 months ago

Selected Answer: A

Yes , agree with A
upvoted 2 times

cegama543 1 year, 9 months ago

Selected Answer: A

agree with that
upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #402 *Topic 1*

A company needs to ingest and handle large amounts of streaming data that its application generates. The application runs on Amazon EC2 instances and sends data to Amazon Kinesis Data Streams, which is configured with default settings. Every other day, the application consumes the data and writes the data to an Amazon S3 bucket for business intelligence (BI) processing. The company observes that Amazon S3 is not receiving all the data that the application sends to Kinesis Data Streams.

What should a solutions architect do to resolve this issue?

A. Update the Kinesis Data Streams default settings by modifying the data retention period. **Most Voted**

B. Update the application to use the Kinesis Producer Library (KPL) to send the data to Kinesis Data Streams.

C. Update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams.

D. Turn on S3 Versioning within the S3 bucket to preserve every version of every object that is ingested in the S3 bucket.

Correct Answer: *A*

Community vote distribution

A (70%) C (27%) Othe

Comments

Wherecanlstart Highly Voted 2 years, 2 months ago

Selected Answer: A

"A Kinesis data stream stores records from 24 hours by default, up to 8760 hours (365 days)."
https://docs.aws.amazon.com/streams/latest/dev/kinesis-extended-retention.html

The question mentioned Kinesis data stream default settings and "every other day". After 24hrs, the data isn't in the Data stream if the default settings is not modified to store data more than 24hrs.

upvoted 33 times

babayomi 7 months, 1 week ago

Thank you for the link

upvoted 2 times

cegama543 Highly Voted 2 years, 3 months ago

Selected Answer: C

C. Update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams.

The best option is to update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams. Kinesis Data Streams scales horizontally by increasing or decreasing the number of shards, which controls the throughput capacity of the stream. By increasing the number of shards, the application will be able to send more data to Kinesis Data Streams, which can help ensure that S3 receives all the data.

upvoted 17 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Answer C:
C. Update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams.

- Answer C updates the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams. By increasing the number of shards, the data is distributed across multiple shards, which allows for increased throughput and ensures that all data is ingested and processed by Kinesis Data Streams.
- Monitoring the Kinesis Data Streams and adjusting the number of shards as needed to handle changes in data throughput can ensure that the application can handle large amounts of streaming data.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 2 months ago

@cegama543, my apologies. Moderator if you can disapprove of the post above? I made a mistake. It is supposed to be intended on the post that I submitted.

Thanks.

upvoted 2 times

CapJackSparrow 2 years, 2 months ago

lets say you had infinity shards... if the retention period is 24 hours and you get the data every 48 hours, you will lose 24 hours of data no matter the amount of shards no?

upvoted 15 times

enzomv 2 years, 2 months ago

Amazon Kinesis Data Streams supports changes to the data record retention period of your data stream. A Kinesis data stream is an ordered sequence of data records meant to be written to and read from in real time. Data records are therefore stored in shards in your stream temporarily. The time period from when a record is added to when it is no longer accessible is called the retention period. A Kinesis data stream stores records from 24 hours by default, up to 8760 hours (365 days).

upvoted 6 times

Rcosmos Most Recent 4 months, 2 weeks ago

Selected Answer: U

C. Atualizar o número de fragmentos do Kinesis:

Aumentar o número de fragmentos melhora a capacidade de processamento e a taxa de transferência do stream. No entanto, este não é o problema aqui, pois a perda de dados ocorre devido ao período de retenção insuficiente, e não devido a limitações de capacidade.

upvoted 1 times

abriggy 10 months, 2 weeks ago

Selected Answer: C

Answer is C

Issue with A) Update the Kinesis Data Streams default settings by modifying the data retention period. is below

Limitation: Modifying the data retention period affects how long data is kept in the stream, but it does not address the issue of the stream's capacity to ingest data. If the stream is unable to handle the incoming data volume, extending the retention period will not resolve the data loss issue.

upvoted 1 times

awsgeek75 1 year, 4 months ago

Selected Answer: A

Every other day, = 48 hours
Default settings = 24 hours

B: Development library so won't help
C: More shards may retain more data but they will have same limitation of 24 hours retention
D: Irrelevant

A: Increase the default limit from 24 hours to 48 hours
upvoted 6 times

pentium75 1 year, 5 months ago

Selected Answer: A

"Default settings" = 24 hour retention
upvoted 5 times

Murtadhaceit 1 year, 6 months ago

Selected Answer: A

KDS has two modes:
1. Provisioned Mode: Answer C would be correct if KDS runs in this mode. We need to increase the number of shards.
2. On-Demand: Scales automatically, which means it doesn't need to adjust the number of shards based on observed throughput.

And since the question does not mention which type, I would go with On-demand. Therefore, A is the correct answer.
upvoted 3 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: A

Data records are stored in shards in a kinesis data stream temporarily. The time period from when a record is added, to when it is no longer accessible is called the retention period. This time period is 24 hours by default, but could be adjusted to 365 days.
Kinesis Data Streams automatically scales the number of shards in response to changes in data volume and traffic, so this rules out option C.

https://docs.aws.amazon.com/streams/latest/dev/service-sizes-and-limits.html#:~:text=the%20number%20of-,shards,-in%20response%20to
upvoted 2 times

Ramdi1 1 year, 8 months ago

Selected Answer: A

I have only voted A because it mentions the default setting in Kinesis, if it did not mention that then I would look to increase the Shards. By default it is 24 hours and can go to 365 days. I think the question should be rephrased slightly. I had trouble deciding between A & C. Also apparently the most voted answer is the correct answer as per some advice I was given.
upvoted 3 times

BrijMohan08 1 year, 9 months ago

Selected Answer: A

Default retention is 24 hrs, but the data read is every other day, so the S3 will never receive the data, Change the default retention period to 48 hours.
upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

By default, a Kinesis data stream is created with one shard. If the data throughput to the stream is higher than the capacity of the single shard, the data stream may not be able to handle all the incoming data, and some data may be lost.
Therefore, to handle the high volume of data that the application sends to Kinesis Data Streams, the number of Kinesis shards should be increased to handle the required throughput.
Kinesis Data Streams shards are the basic units of scalability and availability. Each shard can process up to 1,000 records per second with a maximum of 1 MB of data per second. If the application is sending more data to Kinesis Data Streams than the shards can handle, then some of the data will be dropped.
upvoted 1 times

Guru4Cloud 1 year, 9 months ago

If you have doubts, Please read about Kinesis Data Streams shards.
Ans: A is not the correct answer here
upvoted 1 times

Amycert 1 year, 10 months ago

Selected Answer: A

the default retention period is 24 hours "The default retention period of 24 hours covers scenarios where intermittent lags in processing require catch-up with the real-time data. "
so we should increment this

upvoted 2 times

hsinchang 1 year, 10 months ago

Selected Answer: A

As "Default settings" is mentioned here, I vote for A.

upvoted 2 times

jaydesai8 1 year, 11 months ago

Selected Answer: A

keyword here is - default settings and every other day and since "A Kinesis data stream stores records from 24 hours by default, up to 8760 hours (365 days)."
https://docs.aws.amazon.com/streams/latest/dev/kinesis-extended-retention.html

Will go with A

upvoted 2 times

jayce5 2 years ago

Selected Answer: A

C is wrong because even if you update the number of Kinesis shards, you still need to change the default data retention period first. Otherwise, you would lose data after 24 hours.

upvoted 3 times

antropaws 2 years ago

Selected Answer: C

A is unrelated to the issue. The correct answer is C.

upvoted 1 times

omoakin 2 years ago

Correct Ans. is B

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #403 *Topic 1*

A developer has an application that uses an AWS Lambda function to upload files to Amazon S3 and needs the required permissions to perform the task. The developer already has an IAM user with valid IAM credentials required for Amazon S3.

What should a solutions architect do to grant the permissions?

A. Add required IAM permissions in the resource policy of the Lambda function.

B. Create a signed request using the existing IAM credentials in the Lambda function.

C. Create a new IAM user and use the existing IAM credentials in the Lambda function.

D. Create an IAM execution role with the required permissions and attach the IAM role to the Lambda function. Most Voted

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 2 months ago

To grant the necessary permissions to an AWS Lambda function to upload files to Amazon S3, a solutions architect should create an IAM execution role with the required permissions and attach the IAM role to the Lambda function. This approach follows the principle of least privilege and ensures that the Lambda function can only access the resources it needs to perform its specific task.

Therefore, the correct answer is D. Create an IAM execution role with the required permissions and attach the IAM role to the Lambda function.

upvoted 7 times

AWSSURI 9 months, 2 weeks ago

Oh you're here

upvoted 3 times

Guru4Cloud Most Recent 1 year, 9 months ago

Selected Answer: D

Create Lambda execution role and attach existing S3 IAM role to the lambda function

upvoted 4 times

Bilalglg93350 2 years, 2 months ago

D. Créez un rôle d'exécution IAM avec les autorisations requises et attachez le rôle IAM à la fonction Lambda.

L'architecte de solutions doit créer un rôle d'exécution IAM ayant les autorisations nécessaires pour accéder à Amazon S3 et effectuer les opérations requises (par exemple, charger des fichiers). Ensuite, le rôle doit être associé à la fonction Lambda, de sorte que la fonction puisse assumer ce rôle et avoir les autorisations nécessaires pour interagir avec Amazon S3.

upvoted 4 times

nileshlg 2 years, 2 months ago

Selected Answer: D

Answer is D

upvoted 3 times

kampatra 2 years, 2 months ago

Selected Answer: D

D - correct ans

upvoted 3 times

sitha 2 years, 2 months ago

Selected Answer: D

Create Lambda execution role and attach existing S3 IAM role to the lambda function

upvoted 3 times

ktulu2602 2 years, 2 months ago

Selected Answer: D

Definitely D

upvoted 3 times

Nithin1119 2 years, 3 months ago

Selected Answer: D

ddddddd

upvoted 2 times

[Removed] 2 years, 3 months ago

Selected Answer: D

dddddddd

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #404 *Topic 1*

A company has deployed a serverless application that invokes an AWS Lambda function when new documents are uploaded to an Amazon S3 bucket. The application uses the Lambda function to process the documents. After a recent marketing campaign, the company noticed that the application did not process many of the documents.

What should a solutions architect do to improve the architecture of this application?

A. Set the Lambda function's runtime timeout value to 15 minutes.

B. Configure an S3 bucket replication policy. Stage the documents in the S3 bucket for later processing.

C. Deploy an additional Lambda function. Load balance the processing of the documents across the two Lambda functions.

D. Create an Amazon Simple Queue Service (Amazon SQS) queue. Send the requests to the queue. Configure the queue as an event source for Lambda. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 2 months ago

Selected Answer: D

To improve the architecture of this application, the best solution would be to use Amazon Simple Queue Service (Amazon SQS) to buffer the requests and decouple the S3 bucket from the Lambda function. This will ensure that the documents are not lost and can be processed at a later time if the Lambda function is not available.

This will ensure that the documents are not lost and can be processed at a later time if the Lambda function is not available. By using Amazon SQS, the architecture is decoupled and the Lambda function can process the documents in a scalable and fault-tolerant manner.

upvoted 5 times

Guru4Cloud Most Recent 1 year, 9 months ago

Selected Answer: D

D. Create an Amazon Simple Queue Service (Amazon SQS) queue. Send the requests to the queue. Configure the queue as an event source for Lambd

upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: D

D is the best approach

upvoted 3 times

Russs99 2 years, 2 months ago

Selected Answer: D

D is the correct answer

upvoted 3 times

Bilalglg93350 2 years, 2 months ago

D. Créez une file d’attente Amazon Simple Queue Service (Amazon SQS). Envoyez les demandes à la file d’attente. Configurez la file d’attente en tant que source d’événement pour Lambda.

Cette solution permet de gérer efficacement les pics de charge et d'éviter la perte de documents en cas d'augmentation soudaine du trafic. Lorsque de nouveaux documents sont chargés dans le compartiment Amazon S3, les demandes sont envoyées à la file d'attente Amazon SQS, qui agit comme un tampon. La fonction Lambda est déclenchée en fonction des événements dans la file d'attente, ce qui permet un traitement équilibré et évite que l'application ne soit submergée par un grand nombre de documents simultanés.

upvoted 1 times

JA2018 6 months, 2 weeks ago

Frrom Google Translate:

Create an Amazon Simple Queue Service (Amazon SQS) queue. Send requests to the queue. Configure the queue as an event source for Lambda.

This solution makes it possible to effectively manage load peaks and avoid the loss of documents in the event of a sudden increase in traffic. When new documents are uploaded to the Amazon S3 bucket, requests are sent to the Amazon SQS queue, which acts as a buffer. The Lambda function is triggered based on events in the queue, which provides balanced processing and prevents the application from being overwhelmed by a large number of concurrent documents.

upvoted 1 times

Russs99 2 years, 2 months ago

exactement. si je pouvais explique come cela en Francais aussi

upvoted 1 times

WherecanIstart 2 years, 2 months ago

Selected Answer: D

D is the correct answer.

upvoted 1 times

kampatra 2 years, 2 months ago

Selected Answer: D

D is correct

upvoted 1 times

[Removed] 2 years, 2 months ago

Selected Answer: D

D is correct

upvoted 1 times

[Removed] 2 years, 3 months ago

Selected Answer: D

dddddddd

upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #405 *Topic 1*

A solutions architect is designing the architecture for a software demonstration environment. The environment will run on Amazon EC2 instances in an Auto Scaling group behind an Application Load Balancer (ALB). The system will experience significant increases in traffic during working hours but is not required to operate on weekends.

Which combination of actions should the solutions architect take to ensure that the system can scale to meet demand? (Choose two.)

A. Use AWS Auto Scaling to adjust the ALB capacity based on request rate.

B. Use AWS Auto Scaling to scale the capacity of the VPC internet gateway.

C. Launch the EC2 instances in multiple AWS Regions to distribute the load across Regions.

D. Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization. Most Voted

E. Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week. Most Voted

Correct Answer: *DE*

Community vote distribution

DE (70%) AD (15%) Other (15%)

Comments

pentium75 Highly Voted 1 year, 5 months ago

Selected Answer: DE

Not A - "AWS Auto Scaling" cannot adjust "ALB capacity" (https://aws.amazon.com/autoscaling/faqs/)
Not B - VPC internet gateway has nothing to do with this
Not C - Regions have nothing to do with scaling

"The system will experience significant increases in traffic during working hours" -> addressed by D
"But is not required to operate on weekends" -> addressed by E

upvoted 15 times

foha2012 1 year, 4 months ago

Good explanation!

upvoted 2 times

cd93 Highly Voted 1 year, 9 months ago

What does "ALB capacity" even means anyway? It should be "Target Group capacity" no?
Answer should be DE, as D is a more comprehensive answer (and more practical in real life)
upvoted 13 times

bora4motion Most Recent 1 month, 1 week ago

Selected Answer: DE

who on this earth voted a?
upvoted 1 times

BigHammer 1 year, 9 months ago

AD
E - the question doesn't ask about cost. Also, shutting it down during the weekend does nothing to improve scaling during the week. It doesn't address the requirements.
upvoted 2 times

JA2018 6 months, 2 weeks ago

Key in STEM:

Nobody will use the system during weekend
upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: DE

The solutions architect should take actions D and E:

D) Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization. This will allow the Auto Scaling group to dynamically scale in and out based on demand.

E) Use scheduled scaling to change the Auto Scaling group capacity to zero on weekends when traffic is expected to be low. This will minimize costs by terminating unused instances.
upvoted 7 times

fuzzycr 1 year, 10 months ago

Selected Answer: AE

Basado en los requerimientos la opción que se requiere para optimizar los costos de 0 operaciones en los fines de semana
upvoted 2 times

jaydesai8 1 year, 11 months ago

Selected Answer: DE

DE - This seems more close for the auto scaling -
A - Its says auto scaling on ALB, but it should always be on EC2 instances and not ELB
upvoted 7 times

XaviL 1 year, 11 months ago

Hi guys, very simple
* A. because the question are asking abount request rate!!!! This is a requirement!
* E. The weekend is not necessary to execute anything!

A&D. Is not possible, way you can put an ALB capacity based in cpu and in request rate???? You need to select one or another option (and this is for all questions here guys!)
upvoted 3 times

[Removed] 1 year, 11 months ago

Selected Answer: AE

ALBRequestCountPerTarget—Average Application Load Balancer request count per target.
https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scaling-target-tracking.html#target-tracking-choose-metrics

It is possible to set to zero. "is not required to operate on weekends" means the instances are not required during the

weekends.
https://docs.aws.amazon.com/autoscaling/ec2/userguide/asg-capacity-limits.html
upvoted 3 times

pentium75 1 year, 5 months ago

A says to scale "ALB capacity", not number of EC2 instances. But "AWS Auto Scaling" cannot scale ALB capacity.
upvoted 1 times

Uzi_m 2 years ago

Option E is incorrect because the question specifically mentions an increase in traffic during working hours. Therefore, it is not advisable to schedule the instances for 24 hours using default settings throughout the entire week.
E. Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week.
upvoted 1 times

omoakin 2 years ago

AD are the correct answs
upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: ADE

Either one or two or all of these combinations will meet the need:
Use AWS Auto Scaling to adjust the ALB capacity based on request rate.
Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization.
Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week.
upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Scheduled scaling was specifically designed to handle these kind of requirements.
I therefore take out target scaling.
https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-scheduled-scaling.html#:~:text=RSS-,Scheduled%20scaling,-helps%20you%20to
upvoted 2 times

Joe94KR 2 years, 1 month ago

Selected Answer: DE

https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scaling-target-tracking.html#target-tracking-choose-metrics

Based on docs, ASG can't track ALB's request rate, so the answer is D&E
meanwhile ASG can track CPU rates.
upvoted 5 times

[Removed] 2 years ago

The link shows:
ALBRequestCountPerTarget—Average Application Load Balancer request count per target.
upvoted 2 times

kraken21 2 years, 2 months ago

Selected Answer: DE

Scaling should be at the ASG not ALB. So, not sure about "Use AWS Auto Scaling to adjust the ALB capacity based on request rate"
upvoted 6 times

channn 2 years, 2 months ago

Selected Answer: AD

A. Use AWS Auto Scaling to adjust the ALB capacity based on request rate: This will allow the system to scale up or down based on incoming traffic demand. The solutions architect should use AWS Auto Scaling to monitor the request rate and adjust the ALB capacity as needed.

D. Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization: This will allow the system to scale up or down based on the CPU utilization of the EC2 instances in the Auto Scaling group. The solutions architect should use a target tracking scaling policy to maintain a specific CPU utilization target and adjust the number of EC2 instances in the Auto Scaling group accordingly.

upvoted 9 times

pentium75 1 year, 5 months ago

Auto scaling for ALB capacity?

upvoted 3 times

neosis91 2 years, 2 months ago

Selected Answer: AD

A. Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization. This approach allows the Auto Scaling group to automatically adjust the number of instances based on the specified metric, ensuring that the system can scale to meet demand during working hours.

D. Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week. This approach allows the Auto Scaling group to reduce the number of instances to zero during weekends when traffic is expected to be low. It will help the organization to save costs by not paying for instances that are not needed during weekends.

Therefore, options A and D are the correct answers. Options B and C are not relevant to the scenario, and option E is not a scalable solution as it would require manual intervention to adjust the group capacity every week.

upvoted 1 times

zooba72 2 years, 2 months ago

Selected Answer: DE

This is why I don't believe A is correct use auto scaling to adjust the ALB .... D&E

upvoted 4 times

pentium75 1 year, 5 months ago

Autoscaling can't scale the ALB

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #406 *Topic 1*

A solutions architect is designing a two-tiered architecture that includes a public subnet and a database subnet. The web servers in the public subnet must be open to the internet on port 443. The Amazon RDS for MySQL DB instance in the database subnet must be accessible only to the web servers on port 3306.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

A. Create a network ACL for the public subnet. Add a rule to deny outbound traffic to 0.0.0.0/0 on port 3306.

B. Create a security group for the DB instance. Add a rule to allow traffic from the public subnet CIDR block on port 3306.

C. Create a security group for the web servers in the public subnet. Add a rule to allow traffic from 0.0.0.0/0 on port 443. **Most Voted**

D. Create a security group for the DB instance. Add a rule to allow traffic from the web servers' security group on port 3306. **Most Voted**

E. Create a security group for the DB instance. Add a rule to deny all traffic except traffic from the web servers' security group on port 3306.

Correct Answer: *CD*

Community vote distribution

CD (100%)

Comments

Guru4Cloud Highly Voted 1 year, 3 months ago

Selected Answer: CD

Remember guys that SG is not used for Deny action, just Allow
upvoted 11 times

datmd77 Highly Voted 1 year, 7 months ago

Selected Answer: CD

Remember guys that SG is not used for Deny action, just Allow
upvoted 5 times

Tendu Most Recent 3 months, 2 weeks ago

Selected Answer: CD

Selected Answer: CD

C and D

upvoted 1 times

waldirlsantos 7 months, 4 weeks ago

Selected Answer: CD

The following are the default rules for a security group that you create:

Allows no inbound traffic

Allows all outbound traffic

upvoted 3 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: CD

'must be accessible only to the web servers' is the key here.
Option B almost threw me off, but with this then all that exists in the public subnet would be able to access the DB security group.
Therefore C,D well applies the principle of least privilege.

upvoted 5 times

Buruguduystunstugudunstuy 1 year, 8 months ago

Selected Answer: CD

To meet the requirements of allowing access to the web servers in the public subnet on port 443 and the Amazon RDS for MySQL DB instance in the database subnet on port 3306, the best solution would be to create a security group for the web servers and another security group for the DB instance, and then define the appropriate inbound and outbound rules for each security group.

1. Create a security group for the web servers in the public subnet. Add a rule to allow traffic from 0.0.0.0/0 on port 443.
2. Create a security group for the DB instance. Add a rule to allow traffic from the web servers' security group on port 3306.

This will allow the web servers in the public subnet to receive traffic from the internet on port 443, and the Amazon RDS for MySQL DB instance in the database subnet to receive traffic only from the web servers on port 3306.

upvoted 3 times

kampatra 1 year, 8 months ago

Selected Answer: CD

CD - Correct ans.

upvoted 3 times

Eden 1 year, 8 months ago

I choose CE

upvoted 1 times

lili_9 1 year, 8 months ago

CE support @sitha

upvoted 1 times

sitha 1 year, 9 months ago

Answer: CE . The solution is to deny accessing DB from Internet and allow only access from webserver.

upvoted 1 times

KAUS2 1 year, 9 months ago

Selected Answer: CD

C & D are the right choices. correct

upvoted 2 times

KS2020 1 year, 9 months ago

why not CE?

upvoted 2 times

upvoted 2 times

kampatra 1 year, 8 months ago

By default Security Group deny all trafic and we need to configure to enable.

upvoted 5 times

[Removed] 1 year, 8 months ago

Characteristics of security group rules

You can specify allow rules, but not deny rules.
https://docs.aws.amazon.com/vpc/latest/userguide/VPC_SecurityGroups.html

upvoted 3 times

[Removed] 1 year, 9 months ago

Selected Answer: CD

cdcdcdcdcdc

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #407 *Topic 1*

A company is implementing a shared storage solution for a gaming application that is hosted in the AWS Cloud. The company needs the ability to use Lustre clients to access data. The solution must be fully managed.

Which solution meets these requirements?

A. Create an AWS DataSync task that shares the data as a mountable file system. Mount the file system to the application server.

B. Create an AWS Storage Gateway file gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.

C. Create an Amazon Elastic File System (Amazon EFS) file system, and configure it to support Lustre. Attach the file system to the origin server. Connect the application server to the file system.

D. Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system. Most Voted

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 1 year, 2 months ago

Selected Answer: D

To meet the requirements of a shared storage solution for a gaming application that can be accessed using Lustre clients and is fully managed, the best solution would be to use Amazon FSx for Lustre.

Amazon FSx for Lustre is a fully managed file system that is optimized for compute-intensive workloads, such as high-performance computing, machine learning, and gaming. It provides a POSIX-compliant file system that can be accessed using Lustre clients and offers high performance, scalability, and data durability.

This solution provides a highly available, scalable, and fully managed shared storage solution that can be accessed using Lustre clients. Amazon FSx for Lustre is optimized for compute-intensive workloads and provides high performance and durability.

upvoted 6 times

Buruguduystunstugudunstuy 1 year, 2 months ago

Answer A, creating an AWS DataSync task that shares the data as a mountable file system and mounting the file system to the application server, may not provide the required performance and scalability for a gaming application.

Answer B, creating an AWS Storage Gateway file gateway and connecting the application server to the file share, may not provide the required performance and scalability for a gaming application.

Answer C, creating an Amazon Elastic File System (Amazon EFS) file system and configuring it to support Lustre, may not provide the required performance and scalability for a gaming application and may require additional configuration and management overhead.

upvoted 3 times

Manimgh Most Recent 1 month, 2 weeks ago

Selected Answer: D

Lustre = FSx

upvoted 1 times

Tendu 3 months, 2 weeks ago

Selected Answer: D

D is correct answer.

upvoted 1 times

Guru4Cloud 9 months, 2 weeks ago

Selected Answer: D

Lustre clients = Amazon FSx for Lustre file system

upvoted 4 times

TariqKipkemei 1 year ago

Selected Answer: D

Lustre clients = Amazon FSx for Lustre file system

upvoted 4 times

kampatra 1 year, 2 months ago

Selected Answer: D

D - correct ans

upvoted 2 times

kprakashbehera 1 year, 2 months ago

Selected Answer: D

FSx for Lustre
DDDDDD

upvoted 2 times

KAUS2 1 year, 2 months ago

Selected Answer: D

Amazon FSx for Lustre is the right answer
• Lustre is a type of parallel distributed file system, for large-scale computing, Machine Learning, High Performance Computing (HPC)
• Video Processing, Financial Modeling, Electronic Design Automatio

upvoted 2 times

cegama543 1 year, 3 months ago

Selected Answer: D

Option D is the best solution because Amazon FSx for Lustre is a fully managed, high-performance file system that is designed to support compute-intensive workloads, such as those required by gaming applications. FSx for Lustre provides sub-millisecond access to petabyte-scale file systems, and supports Lustre clients natively. This means that the gaming application can access the shared data directly from the FSx for Lustre file system without the need for additional configuration or setup.

Additionally, FSx for Lustre is a fully managed service, meaning that AWS takes care of all maintenance, updates, and patches for the file system, which reduces the operational overhead required by the company.

upvoted 2 times

upvoted 2 times

[Removed] 1 year, 3 months ago

Selected Answer: D

dddddddddddd

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #408 *Topic 1*

A company runs an application that receives data from thousands of geographically dispersed remote devices that use UDP. The application processes the data immediately and sends a message back to the device if necessary. No data is stored.

The company needs a solution that minimizes latency for the data transmission from the devices. The solution also must provide rapid failover to another AWS Region.

Which solution will meet these requirements?

A. Configure an Amazon Route 53 failover routing policy. Create a Network Load Balancer (NLB) in each of the two Regions. Configure the NLB to invoke an AWS Lambda function to process the data.

B. Use AWS Global Accelerator. Create a Network Load Balancer (NLB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the NLProcess the data in Amazon ECS. **Most Voted**

C. Use AWS Global Accelerator. Create an Application Load Balancer (ALB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the ALB. Process the data in Amazon ECS.

D. Configure an Amazon Route 53 failover routing policy. Create an Application Load Balancer (ALB) in each of the two Regions. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the ALB. Process the data in Amazon ECS.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

UnluckyDucky Highly Voted 1 year, 8 months ago

Selected Answer: B

Key words: geographically dispersed, UDP.

Geographically dispersed (related to UDP) - Global Accelerator - multiple entrances worldwide to the AWS network to provide better transfer rates.
UDP - NLB (Network Load Balancer)

UDP - NLB (Network Load Balancer).
upvoted 15 times

Ruhi02 Highly Voted 1 year, 9 months ago

Answer should be B.. there is typo mistake in B. Correct Answer is : Use AWS Global Accelerator. Create a Network Load Balancer (NLB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the NLB. Process the data in Amazon ECS.
upvoted 5 times

wizcloudifa Most Recent 7 months, 3 weeks ago

Selected Answer: B

if its UDP it has to be Global Accelator + NLB package, plus it has the provision for rapid failover as well, piece of cake.
upvoted 3 times

sandordini 7 months, 3 weeks ago

Selected Answer: B

UDP: NLB + AWS Global Accelerator
upvoted 3 times

zinabu 8 months, 1 week ago

UDP/TCP=NLB
rapid failover= AWS global accelerator
upvoted 3 times

ferdzcruz 11 months, 1 week ago

devices that use UDP = NLB
upvoted 2 times

ferdzcruz 11 months, 1 week ago

minimizes latency = AWS Global Accelerator
upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: B

This option meets the requirements:

Global Accelerator provides UDP support and minimizes latency using the AWS global network.
Using NLBs allows the UDP traffic to be load balanced across Availability Zones.
ECS Fargate provides rapid scaling and failover across Regions.
NLB endpoints allow rapid failover if one Region goes down.
upvoted 2 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: B

UDP = AWS Global Accelerator and Network Load Balancer
upvoted 2 times

kraken21 1 year, 8 months ago

Selected Answer: B

Global accelerator for multi region automatic failover. NLB for UDP.
upvoted 3 times

MaxMa 1 year, 8 months ago

why not A?
upvoted 1 times

kraken21 1 year, 8 months ago

NLBs do not support lambda target type. Tricky!!! https://docs.aws.amazon.com/elasticloadbalancing/latest/network/load-balancer-target-groups.html

upvoted 10 times

Buruguduystunstugudunstuy 1 year, 8 months ago

Selected Answer: B

To meet the requirements of minimizing latency for data transmission from the devices and providing rapid failover to another AWS Region, the best solution would be to use AWS Global Accelerator in combination with a Network Load Balancer (NLB) and Amazon Elastic Container Service (Amazon ECS).

AWS Global Accelerator is a service that improves the availability and performance of applications by using static IP addresses (Anycast) to route traffic to optimal AWS endpoints. With Global Accelerator, you can direct traffic to multiple Regions and endpoints, and provide automatic failover to another AWS Region.

upvoted 4 times

[Removed] 1 year, 9 months ago

Selected Answer: B

bbbbbbbb

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #409 *Topic 1*

A solutions architect must migrate a Windows Internet Information Services (IIS) web application to AWS. The application currently relies on a file share hosted in the user's on-premises network-attached storage (NAS). The solutions architect has proposed migrating the IIS web servers to Amazon EC2 instances in multiple Availability Zones that are connected to the storage solution, and configuring an Elastic Load Balancer attached to the instances.

Which replacement to the on-premises file share is MOST resilient and durable?

A. Migrate the file share to Amazon RDS.

B. Migrate the file share to AWS Storage Gateway.

C. Migrate the file share to Amazon FSx for Windows File Server. **Most Voted**

D. Migrate the file share to Amazon Elastic File System (Amazon EFS).

Correct Answer: *C*

Community vote distribution

C (95%) D (5%)

Comments

channn Highly Voted 1 year, 8 months ago

Selected Answer: C

A) RDS is a database service
B) Storage Gateway is a hybrid cloud storage service that connects on-premises applications to AWS storage services.
D) provides shared file storage for Linux-based workloads, but it does not natively support Windows-based workloads.

upvoted 9 times

Buruguduystunstugudunstuy Highly Voted 1 year, 8 months ago

Selected Answer: C

The most resilient and durable replacement for the on-premises file share in this scenario would be Amazon FSx for Windows File Server.

Amazon FSx is a fully managed Windows file system service that is built on Windows Server and provides native support for the SMB protocol. It is designed to be highly available and durable, with built-in backup and restore capabilities. It is also fully integrated with AWS security services, providing encryption at rest and in transit, and it can be configured to meet compliance standards.

upvoted 7 times

Buruguduystunstugudunstuy 1 year, 8 months ago

Migrating the file share to Amazon RDS or AWS Storage Gateway is not appropriate as these services are designed for database workloads and block storage respectively, and do not provide native support for the SMB protocol.

Migrating the file share to Amazon EFS (Linux ONLY) could be an option, but Amazon FSx for Windows File Server would be more appropriate in this case because it is specifically designed for Windows file shares and provides better performance for Windows applications.

upvoted 6 times

Manimgh Most Recent 1 month, 2 weeks ago

Selected Answer: C

IIS = FSx

upvoted 1 times

com7 1 year ago

Selected Answer: C

Windows Server to FSx For Windows

upvoted 3 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: C

Windows client = Amazon FSx for Windows File Serve

upvoted 2 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: C

Windows client = Amazon FSx for Windows File Server

upvoted 3 times

Grace83 1 year, 8 months ago

Obviously C is the correct answer - FSx for Windows - Windows

upvoted 5 times

UnluckyDucky 1 year, 8 months ago

Selected Answer: C

FSx for Windows - Windows.
EFS - Linux.

upvoted 6 times

mwwt2022 11 months, 1 week ago

good summary

upvoted 3 times

elearningtakai 1 year, 8 months ago

Selected Answer: D

Amazon EFS is a scalable and fully-managed file storage service that is designed to provide high availability and durability. It can be accessed by multiple EC2 instances across multiple Availability Zones simultaneously. Additionally, it offers automatic and instantaneous data replication across different availability zones within a region, which makes it resilient to failures.

upvoted 2 times

asoli 1 year, 8 months ago

EFS is a wrong choice because it can only work with Linux instances. That application has a Windows web server , so its OS is Windows and EFS cannot connect to it

upvoted 5 times

[Removed] 1 year, 8 months ago

Selected Answer: C

Amazon FSx

upvoted 1 times

sitha 1 year, 9 months ago

Amazon FSx makes it easy and cost effective to launch, run, and scale feature-rich, high-performance file systems in the cloud.
Answer : C

upvoted 2 times

KAUS2 1 year, 9 months ago

Selected Answer: C

FSx for Windows is a fully managed Windows file system share drive . Hence C is the correct answer.

upvoted 3 times

Ruhi02 1 year, 9 months ago

FSx for Windows is ideal in this case. So answer is C.

upvoted 2 times

[Removed] 1 year, 9 months ago

Selected Answer: C

ccccccccc

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #410 *Topic 1*

A company is deploying a new application on Amazon EC2 instances. The application writes data to Amazon Elastic Block Store (Amazon EBS) volumes. The company needs to ensure that all data that is written to the EBS volumes is encrypted at rest.

Which solution will meet this requirement?

A. Create an IAM role that specifies EBS encryption. Attach the role to the EC2 instances.

B. Create the EBS volumes as encrypted volumes. Attach the EBS volumes to the EC2 instances. **Most Voted**

C. Create an EC2 instance tag that has a key of Encrypt and a value of True. Tag all instances that require encryption at the EBS level.

D. Create an AWS Key Management Service (AWS KMS) key policy that enforces EBS encryption in the account. Ensure that the key policy is active.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 1 year, 2 months ago

Selected Answer: B

The solution that will meet the requirement of ensuring that all data that is written to the EBS volumes is encrypted at rest is B. Create the EBS volumes as encrypted volumes and attach the encrypted EBS volumes to the EC2 instances.

When you create an EBS volume, you can specify whether to encrypt the volume. If you choose to encrypt the volume, all data written to the volume is automatically encrypted at rest using AWS-managed keys. You can also use customer-managed keys (CMKs) stored in AWS KMS to encrypt and protect your EBS volumes. You can create encrypted EBS volumes and attach them to EC2 instances to ensure that all data written to the volumes is encrypted at rest.

Answer A is incorrect because attaching an IAM role to the EC2 instances does not automatically encrypt the EBS volumes.

Answer C is incorrect because adding an EC2 instance tag does not ensure that the EBS volumes are encrypted.

upvoted 12 times

Kds53829 Most Recent 7 months, 1 week ago

B is the answer

upvoted 2 times

Guru4Cloud 9 months, 2 weeks ago

Selected Answer: B

B. Create the EBS volumes as encrypted volumes. Attach the EBS volumes to the EC2 instances.
upvoted 2 times

TariqKipkemei 1 year ago

Selected Answer: B

Windows client = Amazon FSx for Windows File Server
upvoted 2 times

TariqKipkemei 7 months, 2 weeks ago

ignore this, mind stuck on last question hhhhhh.
Just create the EBS volumes as encrypted volumes then attach the EBS volumes to the EC2 instances.
upvoted 5 times

elearningtakai 1 year, 2 months ago

Selected Answer: B

The other options either do not meet the requirement of encrypting data at rest (A and C) or do so in a more complex or less efficient manner (D).
upvoted 2 times

Bofi 1 year, 2 months ago

Why not D, EBS encryption require the use of KMS key
upvoted 1 times

Buruguduystunstugudunstuy 1 year, 2 months ago

Answer D is incorrect because creating a KMS key policy that enforces EBS encryption does not automatically encrypt EBS volumes. You need to create encrypted EBS volumes and attach them to EC2 instances to ensure that all data written to the volumes are encrypted at rest.
upvoted 10 times

WherecanIstart 1 year, 2 months ago

Selected Answer: B

Create encrypted EBS volumes and attach encrypted EBS volumes to EC2 instances..
upvoted 3 times

sitha 1 year, 2 months ago

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.Select KMS Keys either default or custom
upvoted 2 times

Ruhi02 1 year, 3 months ago

Answer B. You can enable encryption for EBS volumes while creating them.
upvoted 2 times

[Removed] 1 year, 3 months ago

Selected Answer: B

bbbbbbbb
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #411 *Topic 1*

A company has a web application with sporadic usage patterns. There is heavy usage at the beginning of each month, moderate usage at the start of each week, and unpredictable usage during the week. The application consists of a web server and a MySQL database server running inside the data center. The company would like to move the application to the AWS Cloud, and needs to select a cost-effective database platform that will not require database modifications.

Which solution will meet these requirements?

A. Amazon DynamoDB

B. Amazon RDS for MySQL

C. MySQL-compatible Amazon Aurora Serverless **Most Voted**

D. MySQL deployed on Amazon EC2 in an Auto Scaling group

Correct Answer: *C*

Community vote distribution

C (92%) B (8%)

Comments

channn Highly Voted 1 year, 8 months ago

Selected Answer: C

C: Aurora Serverless is a MySQL-compatible relational database engine that automatically scales compute and memory resources based on application usage. no upfront costs or commitments required.
A: DynamoDB is a NoSQL
B: Fixed cost on RDS class
D: More operation requires

upvoted 12 times

Buruguduystunstugudunstuy Highly Voted 1 year, 8 months ago

Selected Answer: C

Answer C, MySQL-compatible Amazon Aurora Serverless, would be the best solution to meet the company's requirements.

Aurora Serverless can be a cost-effective option for databases with sporadic or unpredictable usage patterns since it automatically scales up or down based on the current workload. Additionally, Aurora Serverless is compatible with MySQL, so it does not require any modifications to the application's database code.

upvoted 6 times

TariqKipkemei Most Recent 1 year, 1 month ago

Selected Answer: C

The is a huge demand for auto-scaling which Amazon RDS cannot do. This contributes to the cost savings as Aurora serverless would scale done in low peak times, this contributes to low costs.

upvoted 4 times

JKevin778 1 year, 2 months ago

Selected Answer: B

RDS is cheaper than Aurora.

upvoted 1 times

pentium75 11 months, 1 week ago

RDS is cheaper than Aurora if you have a fixed instance size, but NOT if you have "unpredictable" usage patterns, then Aurora Serverless (!) is cheaper.

upvoted 6 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: C

Answer C, MySQL-compatible Amazon Aurora Serverless, would be the best solution to meet the company's requirements.

upvoted 2 times

MrAWSAssociate 1 year, 5 months ago

Selected Answer: C

Since we have sporadic & unpredictable usage for DB, Aurora Serverless would be fit more cost-efficient for this case scenario than RDS MySQL. https://www.techtarget.com/searchcloudcomputing/answer/When-should-I-use-Amazon-RDS-vs-Aurora-Serverless

upvoted 1 times

antropaws 1 year, 6 months ago

Selected Answer: C

C for sure.

upvoted 3 times

klayytech 1 year, 8 months ago

Selected Answer: B

Amazon RDS for MySQL is a cost-effective database platform that will not require database modifications. It makes it easier to set up, operate, and scale MySQL deployments in the cloud. With Amazon RDS, you can deploy scalable MySQL servers in minutes with cost-efficient and resizable hardware capacity[2].

Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability. DynamoDB is a good choice for applications that require low-latency data access[1].

MySQL-compatible Amazon Aurora Serverless is an on-demand, auto-scaling configuration for Amazon Aurora (MySQL-compatible edition), where the database will automatically start up, shut down, and scale capacity up or down based on your application's needs[3].

So, Amazon RDS for MySQL is the best option for your requirements.

upvoted 2 times

klayytech 1 year, 8 months ago

sorry i will change to C , because

Amazon RDS for MySQL is a fully-managed relational database service that makes it easy to set up, operate, and scale MySQL deployments in the cloud. Amazon Aurora Serverless is an on-demand, auto-scaling configuration for Amazon Aurora (MySQL-compatible edition), where the database will automatically start up, shut down, and scale capacity up or down based on your application’s needs. It is a simple, cost-effective option for infrequent, intermittent, or unpredictable workloads.

upvoted 3 times

boxu03 1 year, 8 months ago

Selected Answer: C

Amazon Aurora Serverless : a simple, cost-effective option for infrequent, intermittent, or unpredictable workloads

upvoted 4 times

[Removed] 1 year, 9 months ago

Selected Answer: C

cccccccccccccccccccc

upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #412 *Topic 1*

An image-hosting company stores its objects in Amazon S3 buckets. The company wants to avoid accidental exposure of the objects in the S3 buckets to the public. All S3 objects in the entire AWS account need to remain private.

Which solution will meet these requirements?

A. Use Amazon GuardDuty to monitor S3 bucket policies. Create an automatic remediation action rule that uses an AWS Lambda function to remediate any change that makes the objects public.

B. Use AWS Trusted Advisor to find publicly accessible S3 buckets. Configure email notifications in Trusted Advisor when a change is detected. Manually change the S3 bucket policy if it allows public access.

C. Use AWS Resource Access Manager to find publicly accessible S3 buckets. Use Amazon Simple Notification Service (Amazon SNS) to invoke an AWS Lambda function when a change is detected. Deploy a Lambda function that programmatically remediates the change.

D. Use the S3 Block Public Access feature on the account level. Use AWS Organizations to create a service control policy (SCP) that prevents IAM users from changing the setting. Apply the SCP to the account. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (96%) A

Comments

Ruhi02 Highly Voted 2 years, 3 months ago

Answer is D ladies and gentlemen. While guard duty helps to monitor s3 for potential threats its a reactive action. We should always be proactive and not reactive in our solutions so D, block public access to avoid any possibility of the info becoming publicly accessible

upvoted 23 times

Buruguduystunstugudunstuy Highly Voted 2 years, 2 months ago

Selected Answer: D

Answer D is the correct solution that meets the requirements. The S3 Block Public Access feature allows you to restrict public access to S3 buckets and objects within the account. You can enable this feature at the account level to prevent any S3 bucket from being made public, regardless of the bucket policy settings. AWS Organizations can be used to apply a Service Control Policy (SCP) to the account to prevent IAM users from changing this setting, ensuring that all S3 objects remain private. This is a straightforward and effective solution that requires minimal operational overhead.

straightforward and effective solution that requires minimal operational overhead.

upvoted 10 times

noircesar25 Most Recent 1 year, 3 months ago

its 1 aws account, how could D be the answer?

upvoted 1 times

JA2018 6 months, 2 weeks ago

make no difference, 1 AWS account, 10 AWS accounts or > 100 AWS accounts, SCP enforces the same guard rails

upvoted 2 times

TariqKipkemei 1 year, 7 months ago

Selected Answer: D

Use the S3 Block Public Access feature on the account level. Use AWS Organizations to create a service control policy (SCP) that prevents IAM users from changing the setting. Apply the SCP to the account

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

Use the S3 Block Public Access feature on the account level. Use AWS Organizations to create a service control policy (SCP) that prevents IAM users from changing the setting. Apply the SCP to the account

upvoted 2 times

MrAWSAssociate 1 year, 11 months ago

Selected Answer: A

A is correct!

upvoted 1 times

pentium75 1 year, 5 months ago

No, first it would not remove any existing public access (only detect changes), second it would just detect and then remediate, but in the meantime someone could access the objects. It's clearly D.

upvoted 3 times

Yadav_Sanjay 2 years ago

Selected Answer: D

https://docs.aws.amazon.com/AmazonS3/latest/userguide/access-control-block-public-access.html

upvoted 4 times

elearningtakai 2 years, 2 months ago

Selected Answer: D

This is the most effective solution to meet the requirements.

upvoted 3 times

Bofi 2 years, 2 months ago

Selected Answer: D

Option D provided real solution by using bucket policy to restrict public access. Other options were focus on detection which wasn't what was been asked

upvoted 3 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #413 *Topic 1*

An ecommerce company is experiencing an increase in user traffic. The company's store is deployed on Amazon EC2 instances as a two-tier web application consisting of a web tier and a separate database tier. As traffic increases, the company notices that the architecture is causing significant delays in sending timely marketing and order confirmation email to users. The company wants to reduce the time it spends resolving complex email delivery issues and minimize operational overhead.

What should a solutions architect do to meet these requirements?

A. Create a separate application tier using EC2 instances dedicated to email processing.

B. Configure the web instance to send email through Amazon Simple Email Service (Amazon SES). **Most Voted**

C. Configure the web instance to send email through Amazon Simple Notification Service (Amazon SNS).

D. Create a separate application tier using EC2 instances dedicated to email processing. Place the instances in an Auto Scaling group.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

elearningtakai Highly Voted 1 year, 8 months ago

Selected Answer: B

Amazon SES is a cost-effective and scalable email service that enables businesses to send and receive email using their own email addresses and domains. Configuring the web instance to send email through Amazon SES is a simple and effective solution that can reduce the time spent resolving complex email delivery issues and minimize operational overhead.

upvoted 11 times

Buruguduystunstugudunstuy Highly Voted 1 year, 8 months ago

Selected Answer: B

The best option for addressing the company's needs of minimizing operational overhead and reducing time spent resolving email delivery issues is to use Amazon Simple Email Service (Amazon SES).

Answer A of creating a separate application tier for email processing may add additional complexity to the architecture and require more operational overhead.

Answer C of using Amazon Simple Notification Service (Amazon SNS) is not an appropriate solution for sending marketing and order confirmation emails since Amazon SNS is a messaging service that is designed to send messages to subscribed endpoints or clients.

Answer D of creating a separate application tier using EC2 instances dedicated to email processing placed in an Auto Scaling group is a more complex solution than necessary and may result in additional operational overhead.

upvoted 6 times

waldirlsantos Most Recent 7 months, 4 weeks ago

Selected Answer: B

B meet these requirements

upvoted 2 times

TariqKipkemei 1 year, 1 month ago

Selected Answer: B

Amazon Simple Email Service (Amazon SES) lets you reach customers confidently without an on-premises Simple Mail Transfer Protocol (SMTP) email server using the Amazon SES API or SMTP interface.

upvoted 3 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: B

B. Configure the web instance to send email through Amazon Simple Email Service (Amazon SES)

upvoted 2 times

nileshlg 1 year, 8 months ago

Answer is B

upvoted 2 times

Ruhi02 1 year, 9 months ago

Answer B.. SES is meant for sending high volume e-mail efficiently and securely.
SNS is meant as a channel publisher/subscriber service

upvoted 5 times

[Removed] 1 year, 9 months ago

Selected Answer: B

bbbbbbbb

upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #414 *Topic 1*

A company has a business system that generates hundreds of reports each day. The business system saves the reports to a network share in CSV format. The company needs to store this data in the AWS Cloud in near-real time for analysis.

Which solution will meet these requirements with the LEAST administrative overhead?

A. Use AWS DataSync to transfer the files to Amazon S3. Create a scheduled task that runs at the end of each day.

B. Create an Amazon S3 File Gateway. Update the business system to use a new network share from the S3 File Gateway. **Most Voted**

C. Use AWS DataSync to transfer the files to Amazon S3. Create an application that uses the DataSync API in the automation workflow.

D. Deploy an AWS Transfer for SFTP endpoint. Create a script that checks for new files on the network share and uploads the new files by using SFTP.

Correct Answer: *B*

Community vote distribution

B (87%) Other (13%)

Comments

TariqKipkemei Highly Voted 1 year, 1 month ago

Selected Answer: B

Both Amazon S3 File Gateway and AWS DataSync are suitable for this scenario.
But there is a requirement for 'LEAST administrative overhead'.
Option C involves the creation of an entirely new application to consume the DataSync API, this rules out this option.

upvoted 15 times

channn Highly Voted 1 year, 8 months ago

Selected Answer: B

Key words:
1. near-real-time (A is out)
2. LEAST administrative (C n D is out)

upvoted 11 times

zdi561 Most Recent 4 months, 2 weeks ago

Selected Answer: D

Firstly the requirement is not about hybrid cloud e.g. use the data migrated to aws from on-premise, so B is gone. Secondly it is near real time, A is gone. Thirdly less admistratively, C is one since an application is a headache. D is the best , it only involves a script. SFTP is good for light file transfer. Here there are only hundreds of files in a day.

upvoted 1 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: C

This option has the least administrative overhead because:

Using DataSync avoids having to rewrite the business system to use a new file gateway or SFTP endpoint.
Calling the DataSync API from an application allows automating the data transfer instead of running scheduled tasks or scripts.
DataSync directly transfers files from the network share to S3 without needing an intermediate server

upvoted 2 times

pentium75 11 months, 1 week ago

"Create an application" hell no, the application must run somewhere etc., this is massive "administrative overhead".

upvoted 5 times

antropaws 1 year, 6 months ago

Selected Answer: B

B. Data Sync is better for one time migrations.

upvoted 3 times

kruasan 1 year, 7 months ago

Selected Answer: B

The correct solution here is:

B. Create an Amazon S3 File Gateway. Update the business system to use a new network share from the S3 File Gateway.

This option requires the least administrative overhead because:

- It presents a simple network file share interface that the business system can write to, just like a standard network share. This requires minimal changes to the business system.

- The S3 File Gateway automatically uploads all files written to the share to an S3 bucket in the background. This handles the transfer and upload to S3 without requiring any scheduled tasks, scripts or automation.

- All ongoing management like monitoring, scaling, patching etc. is handled by AWS for the S3 File Gateway.

upvoted 5 times

kruasan 1 year, 7 months ago

The other options would require more ongoing administrative effort:

A) AWS DataSync would require creating and managing scheduled tasks and monitoring them.

C) Using the DataSync API would require developing an application and then managing and monitoring it.

D) The SFTP option would require creating scripts, managing SFTP access and keys, and monitoring the file transfer process.

So overall, the S3 File Gateway requires the least amount of ongoing management and administration as it presents a simple file share interface but handles the upload to S3 in a fully managed fashion. The business system can continue writing to a network share as is, while the files are transparently uploaded to S3.

The S3 File Gateway is the most hands-off, low-maintenance solution in this scenario.

upvoted 4 times

elearningtakai 1 year, 8 months ago

Selected Answer: B

A - creating a scheduled task is not near-real time.
B - The S3 File Gateway caches frequently accessed data locally and automatically uploads it to Amazon S3, providing near-real-time access to the data.

C - creating an application that uses the DataSync API in the automation workflow may provide near-real-time data access, but it requires additional development effort.
D - it requires additional development effort.
upvoted 5 times

zooba72 1 year, 8 months ago

Selected Answer: B

It's B. DataSync has a scheduler and it runs on hour intervals, it cannot be used real-time
upvoted 2 times

Buruguduystunstugudunstuy 1 year, 8 months ago

Selected Answer: C

The correct answer is C. Use AWS DataSync to transfer the files to Amazon S3. Create an application that uses the DataSync API in the automation workflow.

To store the CSV reports generated by the business system in the AWS Cloud in near-real time for analysis, the best solution with the least administrative overhead would be to use AWS DataSync to transfer the files to Amazon S3 and create an application that uses the DataSync API in the automation workflow.

AWS DataSync is a fully managed service that makes it easy to automate and accelerate data transfer between on-premises storage systems and AWS Cloud storage, such as Amazon S3. With DataSync, you can quickly and securely transfer large amounts of data to the AWS Cloud, and you can automate the transfer process using the DataSync API.
upvoted 4 times

Buruguduystunstugudunstuy 1 year, 8 months ago

Answer A, using AWS DataSync to transfer the files to Amazon S3 and creating a scheduled task that runs at the end of each day, is not the best solution because it does not meet the requirement of storing the CSV reports in near-real time for analysis.

Answer B, creating an Amazon S3 File Gateway and updating the business system to use a new network share from the S3 File Gateway, is not the best solution because it requires additional configuration and management overhead.

Answer D, deploying an AWS Transfer for the SFTP endpoint and creating a script to check for new files on the network share and upload the new files using SFTP, is not the best solution because it requires additional scripting and management overhead
upvoted 3 times

COTIT 1 year, 8 months ago

Selected Answer: B

I think B is the better answer, "LEAST administrative overhead"
https://aws.amazon.com/storagegateway/file/?nc1=h_ls
upvoted 5 times

andyto 1 year, 8 months ago

B - S3 File Gateway.
C - this is wrong answer because data migration is scheduled (this is not continuous task), so condition "near-real time" is not fulfilled
upvoted 4 times

Thief 1 year, 8 months ago

C is the best ans
upvoted 1 times

lizzard812 1 year, 8 months ago

Why not A? There is no scheduled job?
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #415 *Topic 1*

A company is storing petabytes of data in Amazon S3 Standard. The data is stored in multiple S3 buckets and is accessed with varying frequency. The company does not know access patterns for all the data. The company needs to implement a solution for each S3 bucket to optimize the cost of S3 usage.

Which solution will meet these requirements with the MOST operational efficiency?

A. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering. **Most Voted**

B. Use the S3 storage class analysis tool to determine the correct tier for each object in the S3 bucket. Move each object to the identified storage tier.

C. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Glacier Instant Retrieval.

D. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 One Zone-Infrequent Access (S3 One Zone-IA).

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

TariqKipkemei Highly Voted 1 year ago
Selected Answer: A

Unknown access patterns for the data = S3 Intelligent-Tiering
upvoted 8 times

channn Highly Voted 1 year, 2 months ago
Selected Answer: A

Key words: 'The company does not know access patterns for all the data', so A.
upvoted 5 times

Guru4Cloud Most Recent 9 months, 2 weeks ago
Selected Answer: A

Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.
upvoted 3 times

Buruguduystunstugudunstuy 1 year, 2 months ago

Selected Answer: A

The correct answer is A.

Creating an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering would be the most efficient solution to optimize the cost of S3 usage. S3 Intelligent-Tiering is a storage class that automatically moves objects between two access tiers (frequent and infrequent) based on changing access patterns. It is a cost-effective solution that does not require any manual intervention to move data to different storage classes, unlike the other options.
upvoted 5 times

Buruguduystunstugudunstuy 1 year, 2 months ago

Answer B, Using the S3 storage class analysis tool to determine the correct tier for each object and manually moving objects to the identified storage tier would be time-consuming and require more operational overhead.

Answer C, Transitioning objects to S3 Glacier Instant Retrieval would be appropriate for data that is accessed less frequently and does not require immediate access.

Answer D, S3 One Zone-IA would be appropriate for data that can be recreated if lost and does not require the durability of S3 Standard or S3 Standard-IA.
upvoted 3 times

COTIT 1 year, 2 months ago

Selected Answer: A

For me is A. Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.

Why?
"S3 Intelligent-Tiering is the ideal storage class for data with unknown, changing, or unpredictable access patterns"
https://aws.amazon.com/s3/storage-classes/intelligent-tiering/
upvoted 3 times

Bofi 1 year, 2 months ago

Selected Answer: A

Once the data traffic is unpredictable, Intelligent-Tiering is the best option
upvoted 3 times

NIL8891 1 year, 2 months ago

Selected Answer: A

Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.
upvoted 2 times

Maximus007 1 year, 2 months ago

Selected Answer: A

A: as exact pattern is not clear
upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #416

Topic 1

A rapidly growing global ecommerce company is hosting its web application on AWS. The web application includes static content and dynamic content. The website stores online transaction processing (OLTP) data in an Amazon RDS database The website's users are experiencing slow page loads.

Which combination of actions should a solutions architect take to resolve this issue? (Choose two.)

A. Configure an Amazon Redshift cluster.

B. Set up an Amazon CloudFront distribution. **Most Voted**

C. Host the dynamic web content in Amazon S3.

D. Create a read replica for the RDS DB instance. **Most Voted**

E. Configure a Multi-AZ deployment for the RDS DB instance.

Correct Answer: *BD*

Community vote distribution

BD (90%) Other (10%)

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 2 months ago

Selected Answer: BD

To resolve the issue of slow page loads for a rapidly growing e-commerce website hosted on AWS, a solutions architect can take the following two actions:

1. Set up an Amazon CloudFront distribution
2. Create a read replica for the RDS DB instance

Configuring an Amazon Redshift cluster is not relevant to this issue since Redshift is a data warehousing service and is typically used for the analytical processing of large amounts of data.

Hosting the dynamic web content in Amazon S3 may not necessarily improve performance since S3 is an object storage service, not a web application server. While S3 can be used to host static web content, it may not be suitable for hosting dynamic web content since S3 doesn't support server-side scripting or processing.

Configuring a Multi-AZ deployment for the RDS DB instance will improve high availability but may not necessarily improve performance.

upvoted 16 times

upvoted 16 times

acts268 Highly Voted 2 years, 2 months ago

Selected Answer: BD

Cloud Front and Read Replica

upvoted 5 times

pentium75 Most Recent 1 year, 5 months ago

Selected Answer: BD

A - Redshift is for OLAP, not OLTP
B - Caching, reduces page load time and server load
C - S3 can't host dynamic (!) content
D - Read Replica is meant for increasing DB performance
E - Multi-AZ is meant for HA (not asked here)

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: BD

The two options that will best help resolve the slow page loads are:

B) Set up an Amazon CloudFront distribution

and

E) Configure a Multi-AZ deployment for the RDS DB instance

Explanation:

CloudFront can cache static content globally and improve latency for static content delivery.
Multi-AZ RDS improves performance and availability of the database driving dynamic content.

upvoted 4 times

MatAlves 8 months, 4 weeks ago

Wrong. Multi-AZ is meant to "enhance availability by deploying a standby instance in a second AZ, and achieve fault tolerance in the event of an AZ or database instance failure."

It does not improve performance.

upvoted 1 times

antropaws 2 years ago

Selected Answer: BD

BD is correct.

upvoted 4 times

TariqKipkemei 2 years ago

Selected Answer: BD

Resolve latency = Amazon CloudFront distribution and read replica for the RDS DB

upvoted 5 times

SamDouk 2 years, 2 months ago

Selected Answer: BD

B and D

upvoted 3 times

klayytech 2 years, 2 months ago

Selected Answer: BD

The website's users are experiencing slow page loads.

To resolve this issue, a solutions architect should take the following two actions:

Create a read replica for the RDS DB instance. This will help to offload read traffic from the primary database instance and

improve performance.
upvoted 3 times

zooba72 2 years, 2 months ago

Selected Answer: BD

Question asked about performance improvements, not HA. Cloudfront & Read Replica
upvoted 3 times

thaotnt 2 years, 2 months ago

Selected Answer: BD

slow page loads. >>> D
upvoted 3 times

andyto 2 years, 2 months ago

Selected Answer: BD

Read Replica will speed up Reads on RDS DB.
E is wrong. It brings HA but doesn't contribute to speed which is impacted in this case. Multi-AZ is Active-Standby solution.
upvoted 1 times

COTIT 2 years, 2 months ago

Selected Answer: BE

I agree with B & E.
B. Set up an Amazon CloudFront distribution. (Amazon CloudFront is a content delivery network (CDN) service)
E. Configure a Multi-AZ deployment for the RDS DB instance. (Good idea for loadbalance the DB workflow)
upvoted 2 times

pentium75 1 year, 5 months ago

Multi-AZ for HA, Read Replica for Scalability

https://aws.amazon.com/rds/features/read-replicas/?nc1=h_ls
upvoted 2 times

Santosh43 2 years, 2 months ago

B and E (as there is nothing mention about read transactions)
upvoted 1 times

awsgeek75 1 year, 4 months ago

Why E? There is nothing mentioned about High Availability also. E is wrong because Multi AZ won't help with scaling
upvoted 2 times

Akademik6 2 years, 2 months ago

Selected Answer: BD

Cloudfront and Read Replica. We don't need HA here.
upvoted 4 times

Bofi 2 years, 2 months ago

Selected Answer: BE

Amazon CloudFront can handle both static and Dynamic contents hence there is not need for option C l.e hosting the static data on Amazon S3. RDS read replica will reduce the amount of reads on the RDS hence leading a better performance. Multi-AZ is for disaster Recovery , which means D is also out.
upvoted 1 times

Thief 2 years, 2 months ago

Selected Answer: BC

CloudFont with S3
upvoted 1 times

pentium75 1 year, 5 months ago

S3 can't host "dynamic content"

upvoted 3 times

NIL8891 2 years, 2 months ago

Selected Answer: BE

B and E

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #417 *Topic 1*

A company uses Amazon EC2 instances and AWS Lambda functions to run its application. The company has VPCs with public subnets and private subnets in its AWS account. The EC2 instances run in a private subnet in one of the VPCs. The Lambda functions need direct network access to the EC2 instances for the application to work.

The application will run for at least 1 year. The company expects the number of Lambda functions that the application uses to increase during that time. The company wants to maximize its savings on all application resources and to keep network latency between the services low.

Which solution will meet these requirements?

A. Purchase an EC2 Instance Savings Plan Optimize the Lambda functions' duration and memory usage and the number of invocations. Connect the Lambda functions to the private subnet that contains the EC2 instances.

B. Purchase an EC2 Instance Savings Plan Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Connect the Lambda functions to a public subnet in the same VPC where the EC2 instances run.

C. Purchase a Compute Savings Plan. Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Connect the Lambda functions to the private subnet that contains the EC2 instances. **Most Voted**

D. Purchase a Compute Savings Plan. Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Keep the Lambda functions in the Lambda service VPC.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 2 months ago

Selected Answer: C

Answer C is the best solution that meets the company's requirements.

By purchasing a Compute Savings Plan, the company can save on the costs of running both EC2 instances and Lambda functions. The Lambda functions can be connected to the private subnet that contains the EC2 instances through a VPC endpoint for AWS services or a VPC peering connection. This provides direct network access to the EC2 instances while keeping the traffic within the private network, which helps to minimize network latency.

Optimizing the Lambda functions' duration, memory usage, number of invocations, and amount of data transferred can help to further minimize costs and improve performance. Additionally, using a private subnet helps to ensure that the EC2 instances are not directly accessible from the public internet, which is a security best practice.

upvoted 18 times

Buruguduystunstugudunstuy 2 years, 2 months ago

Answer A is not the best solution because connecting the Lambda functions directly to the private subnet that contains the EC2 instances may not be scalable as the number of Lambda functions increases. Additionally, using an EC2 Instance Savings Plan may not provide savings on the costs of running Lambda functions.

Answer B is not the best solution because connecting the Lambda functions to a public subnet may not be as secure as connecting them to a private subnet. Also, keeping the EC2 instances in a private subnet helps to ensure that they are not directly accessible from the public internet.

Answer D is not the best solution because keeping the Lambda functions in the Lambda service VPC may not provide direct network access to the EC2 instances, which may impact the performance of the application.

upvoted 9 times

TariqKipkemei Highly Voted 1 year, 7 months ago

Selected Answer: C

Implement Compute Savings Plan because it applies to Lambda usage as well, then connect the Lambda functions to the private subnet that contains the EC2 instances

upvoted 7 times

MatAlves Most Recent 8 months, 4 weeks ago

"Savings Plans are a flexible pricing model that offer low prices on Amazon EC2, AWS Lambda, and AWS Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in $/hour) for a 1 or 3 year term."
- That already excludes A and B.

The question requires to "keep network latency between the services low", which can be achieved by connecting the Lambda functions to the private subnet that contains the EC2 instances.

C is the answer.

upvoted 2 times

[Removed] 12 months ago

C
https://aws.amazon.com/savingsplans/compute-pricing/

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

A Compute Savings Plan covers both EC2 and Lambda and allows maximizing savings on all resources.
Optimizing Lambda configuration reduces costs.
Connecting the Lambda functions to the private subnet with the EC2 instances provides direct network access between them, keeping latency low.
The Lambda functions are isolated in the private subnet rather than public, improving security.

upvoted 4 times

jaehoon090 1 year, 10 months ago

CCCCCCCCCCCCCCCCCCCC

upvoted 2 times

elearningtakai 2 years, 2 months ago

Selected Answer: C

Connect Lambda to Private Subnet contains EC2

upvoted 2 times

zooba72 2 years, 2 months ago

Selected Answer: C

Compute savings plan covers both EC2 & Lambda

upvoted 6 times

Zox42 2 years, 2 months ago

C. I would go with C, because Compute savings plans cover Lambda as well.

upvoted 5 times

andyto 2 years, 2 months ago

A. I would go with A. Saving and low network latency are required.
EC2 instance savings plans offer savings of up to 72%
Compute savings plans offer savings of up to 66%
Placing Lambda on the same private network with EC2 instances provides the lowest latency.

upvoted 1 times

abitwrong 2 years, 2 months ago

EC2 Instance Savings Plans apply to EC2 usage only. Compute Savings Plans apply to usage across Amazon EC2, AWS Lambda, and AWS Fargate. (https://aws.amazon.com/savingsplans/faq/)

Lambda functions need direct network access to the EC2 instances for the application to work and these EC2 instances are in the private subnet. So the correct answer is C.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #418 *Topic 1*

A solutions architect needs to allow team members to access Amazon S3 buckets in two different AWS accounts: a development account and a production account. The team currently has access to S3 buckets in the development account by using unique IAM users that are assigned to an IAM group that has appropriate permissions in the account.

The solutions architect has created an IAM role in the production account. The role has a policy that grants access to an S3 bucket in the production account.

Which solution will meet these requirements while complying with the principle of least privilege?

A. Attach the Administrator Access policy to the development account users.

B. Add the development account as a principal in the trust policy of the role in the production account. Most Voted

C. Turn off the S3 Block Public Access feature on the S3 bucket in the production account.

D. Create a user in the production account with unique credentials for each team member.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

kels1 Highly Voted 1 year, 1 month ago

well, if you made it this far, it means you are persistent :) Good luck with your exam!

upvoted 79 times

SkyZeroZx 1 year, 1 month ago

Thanks good luck for all

upvoted 10 times

Kimnesh 9 months, 3 weeks ago

thank you!

upvoted 6 times

gpt_test Highly Voted 1 year, 2 months ago

Selected Answer: B

By adding the development account as a principal in the trust policy of the IAM role in the production account, you are allowing users from the development account to assume the role in the production account. This allows the team members to access the S3 bucket in the production account without granting them unnecessary privileges.

upvoted 9 times

digital_eye Most Recent 2 months, 1 week ago

Selected Answer: B

Good luck everyone

upvoted 1 times

TariqKipkemei 7 months, 2 weeks ago

Selected Answer: B

Add the development account as a principal in the trust policy of the role in the production account

upvoted 3 times

Guru4Cloud 9 months, 2 weeks ago

Selected Answer: B

The best solution is B) Add the development account as a principal in the trust policy of the role in the production account.

This allows cross-account access to the S3 bucket in the production account by assuming the IAM role. The development account users can assume the role to gain temporary access to the production bucket.

upvoted 5 times

nilandd44gg 11 months, 1 week ago

Selected Answer: B

https://aws.amazon.com/blogs/security/how-to-use-trust-policies-with-iam-roles/

An AWS account accesses another AWS account – This use case is commonly referred to as a cross-account role pattern. It allows human or machine IAM principals from one AWS account to assume this role and act on resources within a second AWS account. A role is assumed to enable this behavior when the resource in the target account doesn't have a resource-based policy that could be used to grant cross-account access.

upvoted 3 times

elearningtakai 1 year, 2 months ago

Selected Answer: B

About Trust policy – The trust policy defines which principals can assume the role, and under which conditions. A trust policy is a specific type of resource-based policy for IAM roles.

Answer A: overhead permission Admin to development.
Answer C: Block public access is a security best practice and seems not relevant to this scenario.
Answer D: difficult to manage and scale

upvoted 3 times

Buruguduystunstugudunstuy 1 year, 2 months ago

Selected Answer: B

Answer A, attaching the Administrator Access policy to development account users, provides too many permissions and violates the principle of least privilege. This would give users more access than they need, which could lead to security issues if their credentials are compromised.

Answer C, turning off the S3 Block Public Access feature, is not a recommended solution as it is a security best practice to enable S3 Block Public Access to prevent accidental public access to S3 buckets.

Answer D, creating a user in the production account with unique credentials for each team member, is also not a recommended solution as it can be difficult to manage and scale for large teams. It is also less secure, as individual user credentials can be more easily compromised.

upvoted 3 times

klayytech 1 year, 2 months ago

Selected Answer: B

The solution that will meet these requirements while complying with the principle of least privilege is to add the development account as a principal in the trust policy of the role in the production account. This will allow team members to access Amazon S3 buckets in two different AWS accounts while complying with the principle of least privilege.

Option A is not recommended because it grants too much access to development account users. Option C is not relevant to this scenario. Option D is not recommended because it does not comply with the principle of least privilege.

upvoted 2 times

Akademik6 1 year, 2 months ago

Selected Answer: B

B is the correct answer

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #419 *Topic 1*

A company uses AWS Organizations with all features enabled and runs multiple Amazon EC2 workloads in the ap-southeast-2 Region. The company has a service control policy (SCP) that prevents any resources from being created in any other Region. A security policy requires the company to encrypt all data at rest.

An audit discovers that employees have created Amazon Elastic Block Store (Amazon EBS) volumes for EC2 instances without encrypting the volumes. The company wants any new EC2 instances that any IAM user or root user launches in ap-southeast-2 to use encrypted EBS volumes. The company wants a solution that will have minimal effect on employees who create EBS volumes.

Which combination of steps will meet these requirements? (Choose two.)

A. In the Amazon EC2 console, select the EBS encryption account attribute and define a default encryption key.

B. Create an IAM permission boundary. Attach the permission boundary to the root organizational unit (OU). Define the boundary to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.

C. Create an SCP. Attach the SCP to the root organizational unit (OU). Define the SCP to deny the ec2:CreateVolume action whenthe ec2:Encrypted condition equals false. **Most Voted**

D. Update the IAM policies for each account to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.

E. In the Organizations management account, specify the Default EBS volume encryption setting. **Most Voted**

Correct Answer: *CE*

Community vote distribution

CE (76%) AE (13%) AC (11%)

Comments

Guru4Cloud Highly Voted 1 year, 3 months ago

Selected Answer: CE

The correct answer is (C) and (E).

Option (C): Creating an SCP and attaching it to the root organizational unit (OU) will deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false. This means that any IAM user or root user in any account in the organization will not

be able to create an EBS volume without encrypting it.
Option (E): Specifying the Default EBS volume encryption setting in the Organizations management account will ensure that all new EBS volumes created in any account in the organization are encrypted by default.

upvoted 12 times

Axaus Highly Voted 1 year, 6 months ago

Selected Answer: CE

CE
Prevent future issues by creating a SCP and set a default encryption.

upvoted 10 times

network_enthusiast Most Recent 1 month, 2 weeks ago

Selected Answer: AC

Answer will be A&C.
AWS does not currently provide a direct setting at the organization level to enforce default EBS encryption across all member accounts in AWS Organizations. However, you can achieve this goal by implementing default encryption at the account level and enforcing compliance using an SCP (Service Control Policy).

upvoted 1 times

Andreew883 1 month, 3 weeks ago

Selected Answer: CE

C and E as shaded by guru4cloud

upvoted 1 times

Jazz888 6 months ago

The problem here is we don't know in which account the workload is on. The account in ap-xx-is that the management account or it's a member account?? That will decide to select either A or E. C is certainly correct

upvoted 2 times

NSA_Poker 6 months, 1 week ago

Selected Answer: CE

(A) is incorrect bc absent of SCP or the Organizations management account, the scope of the EC2 console is too narrow to be applied to 'any IAM user or root user'.

upvoted 2 times

venutadi 7 months, 2 weeks ago

Selected Answer: AC

https://repost.aws/knowledge-center/ebs-automatic-encryption
Newly created Amazon EBS volumes aren't encrypted by default. However, you can turn on default encryption for new EBS volumes and snapshot copies that are created within a specified Region. To turn on encryption by default, use the Amazon Elastic Compute Cloud (Amazon EC2) console.

upvoted 2 times

1rob 11 months ago

Selected Answer: AC

A: will enforce automatic encryption in a account. This will have no effect on employees. Do this in every account.
B: permission boundary is not appropriate here.
C: an SCP will force employees to create encrypted volumes in every account.
D: This would work but is too much maintenance.
E: Setting EBS volume encryption in the Organizations management account will only have impact on volumes in that account, not on other accounts.

upvoted 2 times

pentium75 11 months, 1 week ago

Selected Answer: AE

The solution should "have minimal effect on employees who create EBS volumes". Thus new volumes should automatically be encrypted. Options B, C and D do NOT automatically encrypt volumes, they simply cause requests to create non-encrypted volumes to fail.

upvoted 3 times

dkw2342 9 months ago

IMO the correct solution is AC:

In the Amazon EC2 console, select the EBS encryption account attribute and define a default encryption key.
-> This has to be done in every AWS account separately.

Create an SCP. Attach the SCP to the root organizational unit (OU). Define the SCP to deny the ec2:CreateVolume action whenthe ec2:Encrypted condition equals false.
-> This will just act as a safeguard in case an admin would disable default encryption in the member account, so it should not have any effect on employees who create EBS volumes.

I think an updated question would offer options A and an updated C:

Create an SCP. Attach the SCP to the root organizational unit (OU). Define the SCP to deny the ec2:DisableEbsEncryptionByDefault action.
-> This will prevent disabling default encryption once is has been enabled.

upvoted 2 times

Valder21 1 year, 3 months ago

Wondering if just C would be sufficient?

upvoted 1 times

bjexamprep 1 year, 3 months ago

Seems many people selected E as part of the correct answer. But I didn't find so called Organization level EBS default setting in my Organization management account. I tried setting default EBS encryption setting in my Organization management account, and it didn't apply to the member account. If E cannot guarantee default encryption in all other account, E has no advantage over A. Anyone can explain why E is better than A?

upvoted 4 times

novelai_me 1 year, 5 months ago

Selected Answer: AE

Option A: By default, EBS encryption is not enabled for EC2 instances. However, you can set an EBS encryption by default in your AWS account in the Amazon EC2 console. This ensures that every new EBS volume that is created is encrypted.
Option E: With AWS Organizations, you can centrally set the default EBS encryption for your organization's accounts. This helps in enforcing a consistent encryption policy across your organization.
Option B, C and D are not correct because while you can use IAM policies or SCPs to restrict the creation of unencrypted EBS volumes, this could potentially impact employees' ability to create necessary resources if not properly configured. They might require additional permissions management, which is not mentioned in the requirements. By setting the EBS encryption by default at the account or organization level (Options A and E), you can ensure all new volumes are encrypted without affecting the ability of employees to create resources.

upvoted 3 times

Buruguduystunstugudunstuy 1 year, 5 months ago

Selected Answer: CE

SCPs are a great way to enforce policies across an entire AWS Organization, preventing users from creating resources that do not comply with the set policies.

In AWS Management Console, one can go to EC2 dashboard -> Settings -> Data encryption -> Check "Always encrypt new EBS volumes" and choose a default KMS key. This ensures that every new EBS volume created will be encrypted by default, regardless of how it is created.

upvoted 3 times

PRASAD180 1 year, 6 months ago

1000% CE crt

upvoted 1 times

[Removed] 1 year, 6 months ago

Encryption by default allows you to ensure that all new EBS volumes created in your account are always encrypted, even if you don't specify encrypted=true request parameter.
https://aws.amazon.com/blogs/compute/must-know-best-practices-for-amazon-ebs-encryption/

upvoted 1 times

hiroohiroo 1 year, 6 months ago

Selected Answer: CE

CとEが正しいと考える。
upvoted 3 times

Efren 1 year, 6 months ago

Selected Answer: CE

CE for me as well
upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #420 *Topic 1*

A company wants to use an Amazon RDS for PostgreSQL DB cluster to simplify time-consuming database administrative tasks for production database workloads. The company wants to ensure that its database is highly available and will provide automatic failover support in most scenarios in less than 40 seconds. The company wants to offload reads off of the primary instance and keep costs as low as possible.

Which solution will meet these requirements?

A. Use an Amazon RDS Multi-AZ DB instance deployment. Create one read replica and point the read workload to the read replica.

B. Use an Amazon RDS Multi-AZ DB duster deployment Create two read replicas and point the read workload to the read replicas.

C. Use an Amazon RDS Multi-AZ DB instance deployment. Point the read workload to the secondary instances in the Multi-AZ pair.

D. Use an Amazon RDS Multi-AZ DB cluster deployment Point the read workload to the reader endpoint. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (86%) Other (14%)

Comments

ogerber Highly Voted 2 years ago

Selected Answer: D

A - multi-az instance : failover takes between 60-120 sec
D - multi-az cluster: failover around 35 sec

upvoted 20 times

kelmryan1 1 year, 1 month ago

They want to keep cost low as possible, A is the right answer

upvoted 2 times

MatAlves 8 months, 4 weeks ago

No, the question clearly says "failover support in most scenarios in less than 40 seconds."
D is the only possible answer.
upvoted 2 times

Buruguduystunstugudunstuy Highly Voted 1 year, 12 months ago

Selected Answer: D

The correct answer is:
D. Use an Amazon RDS Multi-AZ DB cluster deployment. Point the read workload to the reader endpoint.

Explanation:
The company wants high availability, automatic failover support in less than 40 seconds, read offloading from the primary instance, and cost-effectiveness.

Answer D is the best choice for several reasons:

1. Amazon RDS Multi-AZ deployments provide high availability and automatic failover support.

2. In a Multi-AZ DB cluster, Amazon RDS automatically provisions and maintains a standby in a different Availability Zone. If a failure occurs, Amazon RDS performs an automatic failover to the standby, minimizing downtime.

3. The "Reader endpoint" for an Amazon RDS DB cluster provides load-balancing support for read-only connections to the DB cluster. Directing read traffic to the reader endpoint helps in offloading read operations from the primary instance.
upvoted 13 times

Kiki_Pass 1 year, 10 months ago

Sorry I'm a bit confused... I thought only Aurora DB Cluster has reader endpoint. Do you by any chance has the link to the doc for RDS Reader Endpoint?
upvoted 3 times

lemur88 1 year, 9 months ago

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/multi-az-db-clusters-concepts-connection-management.html#multi-az-db-clusters-concepts-connection-management-endpoints-reader
upvoted 6 times

zikou Most Recent 9 months, 3 weeks ago

to offload read we use read replicas also there is no such thing as reader endpoint in rds, it is only on aurora
upvoted 2 times

rondelldell 1 year, 1 month ago

Selected Answer: B

https://aws.amazon.com/rds/features/multi-az/
Amazon RDS Multi-AZ with two readable standbys
upvoted 1 times

hro 1 year, 2 months ago

I think the cluster is over-kill - but the company 'wants to use an Amazon RDS ... DB cluster'.
upvoted 3 times

pentium75 1 year, 5 months ago

Selected Answer: D

A would be cheapest but "failover times are typically 60–120 seconds" which does not meet our requirements. We need Multi-AZ DB cluster (not instance). This has a reader endpoint by default, thus no need for additional read replicas (to "keep costs as low as possible").
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Concepts.MultiAZSingleStandby.html
upvoted 7 times

master9 1 year, 5 months ago

Selected Answer: A

in question, it has mentioned that "keep costs as low as possible"

In a Multi-AZ configuration, the DB instances and EBS storage volumes are deployed across two Availability Zones.

In a Multi-AZ configuration, the DB instances and EBS storage volumes are deployed across two Availability Zones.
It provides high availability and failover support for DB instances.
This setup is primarily for disaster recovery.
It involves a primary DB instance and a standby replica, which is a copy of the primary DB instance.
The standby replica is not accessible directly; instead, it serves as a failover target in case the primary instance fails.
upvoted 1 times

potomac 1 year, 7 months ago

Selected Answer: D

It is D.
A is not correct. Multi-AZ DB instance deployment, which creates a primary instance and a standby instance to provide failover support. However, the standby instance does not serve traffic.
upvoted 2 times

maudsha 1 year, 7 months ago

Selected Answer: D

https://aws.amazon.com/blogs/database/choose-the-right-amazon-rds-deployment-option-single-az-instance-multi-az-instance-or-multi-az-database-cluster/#:~:text=Unlike%20Multi%2DAZ%20instance%20deployment,different%20AZs%20serving%20read%20traffic.

According to this the answer is D
"Unlike Multi-AZ instance deployment, where the secondary instance can't be accessed for read or writes, Multi-AZ DB cluster deployment consists of primary instance running in one AZ serving read-write traffic and two other standby running in two different AZs serving read traffic."

You don't have to create read replicas with cluster deployment so B is out.
upvoted 2 times

kwang312 1 year, 8 months ago

D
Fail-over on Multi-AZ DB instance is 60-120s
On Cluster, the time under 35s
upvoted 5 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

D. Use an Amazon RDS Multi-AZ DB cluster deployment. Point the read workload to the reader endpoint
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

Use an Amazon RDS Multi-AZ DB cluster deployment Point the read workload to the reader endpoint.
upvoted 2 times

Eminenza22 1 year, 9 months ago

Selected Answer: A

The solutions architect should use an Amazon RDS Multi-AZ DB instance deployment. The company can create one read replica and point the read workload to the read replica. Amazon RDS provides high availability and failover support for DB instances using Multi-AZ deployments.
upvoted 1 times

Gooniegoogoo 1 year, 11 months ago

and d..

Multi-AZ DB clusters typically have lower write latency when compared to Multi-AZ DB instance deployments. They also allow read-only workloads to run on reader DB instances.
upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: D

This is as case where both option A and D can work, but option D gives 2 DB instances for read compared to only 1 given by option A. Costwise they are the same as both options use 3 DB instances.

upvoted 2 times

Henrytml 2 years ago

Selected Answer: A

lowest cost option, and effective with read replica

upvoted 3 times

antropaws 2 years ago

Selected Answer: D

It's D. Read well: "A company wants to use an Amazon RDS for PostgreSQL DB CLUSTER".

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #421 *Topic 1*

A company runs a highly available SFTP service. The SFTP service uses two Amazon EC2 Linux instances that run with elastic IP addresses to accept traffic from trusted IP sources on the internet. The SFTP service is backed by shared storage that is attached to the instances. User accounts are created and managed as Linux users in the SFTP servers.

The company wants a serverless option that provides high IOPS performance and highly configurable security. The company also wants to maintain control over user permissions.

Which solution will meet these requirements?

A. Create an encrypted Amazon Elastic Block Store (Amazon EBS) volume. Create an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses. Attach the EBS volume to the SFTP service endpoint. Grant users access to the SFTP service.

B. Create an encrypted Amazon Elastic File System (Amazon EFS) volume. Create an AWS Transfer Family SFTP service with elastic IP addresses and a VPC endpoint that has internet-facing access. Attach a security group to the endpoint that allows only trusted IP addresses. Attach the EFS volume to the SFTP service endpoint. Grant users access to the SFTP service. **Most Voted**

C. Create an Amazon S3 bucket with default encryption enabled. Create an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses. Attach the S3 bucket to the SFTP service endpoint. Grant users access to the SFTP service.

D. Create an Amazon S3 bucket with default encryption enabled. Create an AWS Transfer Family SFTP service with a VPC endpoint that has internal access in a private subnet. Attach a security group that allows only trusted IP addresses. Attach the S3 bucket to the SFTP service endpoint. Grant users access to the SFTP service.

Correct Answer: *B*

Community vote distribution

B (85%) Other (15%)

Comments

pentium75 Highly Voted 1 year, 5 months ago

Selected Answer: B

Not A - Transfer Family canj't use EBS
B - Possible and meets requirement
Not C - S3 doesn't guarantee "high IOPS performance"; also there is no "public endpoint that allows only trusted IP addresses" (you can assign a Security Group to a public endpoint but that is not mentioned here)
Not D - Endpoint would be in private subnet, not accessible from Internet at all
upvoted 10 times

alexandercamachop Highly Voted 2 years ago

Selected Answer: B

First Serverless - EFS
Second it says it is attached to the Linux instances at the same time, only EFS can do that.
upvoted 8 times

bora4motion 2 weeks, 2 days ago

dude, if you have linux instances that's no longer serverless =)))
upvoted 1 times

bora4motion Most Recent 2 weeks, 2 days ago

Selected Answer: C

I'm going with C. EFS needs to be attached to something, used with a server/instance.
upvoted 1 times

FlyingHawk 6 months ago

Selected Answer: B

D, the description "VPC endpoint that has internal access in a private subnet" is technically incorrect and doesn't make sense for AWS Transfer Family.
AWS Transfer Family actually supports:

Public endpoints (internet-facing)
VPC endpoints (private network access)

The VPC endpoint doesn't have "internal access" - it's about how the SFTP service is configured to interact with your VPC network resources.
upvoted 1 times

FlyingHawk 6 months ago

A more accurate description for a VPC-based AWS Transfer Family SFTP service would be:

Create an AWS Transfer Family SFTP service with a VPC endpoint
Place the endpoint in a private subnet
Configure security groups to allow only trusted IP sources
Attach the S3 bucket to the SFTP service endpoint

So while the overall strategy in option D (using S3 with VPC endpoint and security group restrictions) could be valid, the specific wording about "internal access" is incorrect.
This technical inaccuracy in the description would make D an incorrect answer in a certification or technical assessment.
upvoted 1 times

FlyingHawk 6 months ago

C public endpoint that allows only trusted IP addresses does not sound technical correct as only the security group can restrict the IP address. the correct description should " Create an AWS Transfer Family SFTP service in public subnet with security group allows only trusted IP address, create the vpc endpoint (gateway endpoint here for saving the cost) of S3 bucket to allow the SFTP service can access it privately, grant user access to the SFTP service via IAM policies and grant the access of S3 to SFTP service via IAM role
upvoted 1 times

FlyingHawk 6 months ago

Selected Answer: C

Amazon S3 provides high throughput and performance suitable for many use cases, including those requiring high IOPS (Input/Output Operations Per Second). S3 is not a block storage solution (like EBS) or a file system (like EFS). While its performance is exceptional for its intended use case (object storage), it may not match the millisecond latency or consistent high IOPS required for transactional databases or other ultra-low latency applications.

For applications where sub-millisecond latency or extremely high random IOPS (e.g., 64,000 IOPS) is required, solutions like EBS or EFS would be better.

upvoted 1 times

FlyingHawk 6 months ago

After read the description of C, I feel the description of "an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses" does not sound correct, the endpoint itself cannot restrict the IP address from internet, only the security group. the correct description should be: Create an AWS Transfer Family SFTP service with a public endpoint and security group that allows only trusted IP addresses. create the vpc endpoint (gateway endpoint here for saving the cost) of S3 bucket to allow the SFTP service can access it privately, grant user access to the SFTP service via IAM policies and grant the access of S3 to SFTP service via IAM role

upvoted 1 times

JA2018 6 months, 2 weeks ago

Selected Answer: B

Actually AWS Transfer Family can use S3, so it's a toss-up between Options B & C but I tend to favour Option B for the following reasons (based on the keys in STEM):

company runs a highly available SFTP service. The SFTP service uses two Amazon EC2 Linux instances that run with elastic IP addresses to accept traffic from trusted IP sources on the internet. The SFTP service is backed by shared storage that is attached to the instances. User accounts are created and managed as Linux users in the SFTP servers.
1. Requires a serverless option that provides high IOPS performance and highly configurable security.

2. User also wants to maintain control over user permissions

upvoted 1 times

523db89 9 months, 3 weeks ago

Option B best meets the company's requirements by leveraging AWS Transfer Family with an EFS volume, ensuring high availability, security, and performance.

upvoted 2 times

NickGordon 1 year, 7 months ago

Selected Answer: B

A is incorrect as EBS is not an option
C is incorrect as when I select public accessible, I don't see an option I can set up trusted IP address
D isi incorrect as it is internal.

B, followed the steps and I can set up a sftp in this way

upvoted 4 times

potomac 1 year, 7 months ago

Selected Answer: B

B
EFS has lower latency and higher throughput than S3 when accessed from within the same availability zone.

upvoted 3 times

thanhnv142 1 year, 7 months ago

C: Because it is server-less. deffinitely not A or B because it utilizes server.

upvoted 1 times

warp 1 year, 7 months ago

Amazon Elastic File System - Serverless, fully elastic file storage:
https://aws.amazon.com/efs/

upvoted 5 times

bsbs1234 1 year, 8 months ago

B,
A), transfer family does not support EBS
C,D), S3 has lower IOPS than EFS

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

Create an encrypted Amazon Elastic File System (Amazon EFS) volume. Create an AWS Transfer Family SFTP service with elastic IP addresses and a VPC endpoint that has internet-facing access. Attach a security group to the endpoint that allows only trusted IP addresses. Attach the EFS volume to the SFTP service endpoint. Grant users access to the SFTP service.

upvoted 2 times

Axeashes 1 year, 12 months ago

https://aws.amazon.com/blogs/storage/use-ip-whitelisting-to-secure-your-aws-transfer-for-sftp-servers/

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: B

EFS is best to serve this purpose.

upvoted 2 times

envest 2 years ago

Answer C (from abylead.com)
Transfer Family offers fully managed serverless support for B2B file transfers via SFTP, AS2, FTPS, & FTP directly in & out of S3 or EFS. For a controlled internet access you can use internet-facing endpts with Transfer SFTP servers & restrict trusted internet sources with VPC's default Sgrp. In addition, S3 Access Points aliases allows you to use S3 bkt names for a unique access control plcy on shared S3 datasets.
Transfer SFTP & S3: https://aws.amazon.com/blogs/apn/how-to-use-aws-transfer-family-to-replace-and-scale-sftp-servers/

A)Transfer SFTP doesn't support EBS, not for share data, & not serverless: infeasible.
B)EFS mounts via ENIs not endpts: infeasible.
D)pub endpt for internet access is missing: infeasible.

upvoted 4 times

omoakin 2 years ago

BBBBBBBBBBBBB

upvoted 1 times

vesen22 2 years ago

Selected Answer: B

EFS all day

upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #422 | *Topic 1*

A company is developing a new machine learning (ML) model solution on AWS. The models are developed as independent microservices that fetch approximately 1 GB of model data from Amazon S3 at startup and load the data into memory. Users access the models through an asynchronous API. Users can send a request or a batch of requests and specify where the results should be sent.

The company provides models to hundreds of users. The usage patterns for the models are irregular. Some models could be unused for days or weeks. Other models could receive batches of thousands of requests at a time.

Which design should a solutions architect recommend to meet these requirements?

A. Direct the requests from the API to a Network Load Balancer (NLB). Deploy the models as AWS Lambda functions that are invoked by the NLB.

B. Direct the requests from the API to an Application Load Balancer (ALB). Deploy the models as Amazon Elastic Container Service (Amazon ECS) services that read from an Amazon Simple Queue Service (Amazon SQS) queue. Use AWS App Mesh to scale the instances of the ECS cluster based on the SQS queue size.

C. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the models as AWS Lambda functions that are invoked by SQS events. Use AWS Auto Scaling to increase the number of vCPUs for the Lambda functions based on the SQS queue size.

D. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the models as Amazon Elastic Container Service (Amazon ECS) services that read from the queue. Enable AWS Auto Scaling on Amazon ECS for both the cluster and copies of the service based on the queue size. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

examtopictempacc Highly Voted 1 year, 6 months ago

asynchronous=SQS, microservices=ECS.
Use AWS Auto Scaling to adjust the number of ECS services.

upvoted 17 times

.

TariqKipkemei 1 year, 6 months ago

good breakdown :)
upvoted 4 times

TariqKipkemei Highly Voted 1 year, 6 months ago

Selected Answer: D

For once examtopic answer is correct :) haha...

Batch requests/async = Amazon SQS
Microservices = Amazon ECS
Workload variations = AWS Auto Scaling on Amazon ECS
upvoted 11 times

wizcloudifa Most Recent 7 months, 3 weeks ago

Selected Answer: D

ALB is mentioned in other options to distract you, you dont need ALB for scaling here, we would need ECS autoscaling, they play with that idea in option B a bit however D gets it in a completely optimized way.... A and C both have lambda which for Machine learning models with workloads on heavy side, will not fly
upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: D

I go with everyone D.
upvoted 3 times

alexandercamachop 1 year, 6 months ago

Selected Answer: D

D, no need for an App Load balancer like C says, no where in the text.
SQS is needed to ensure all request gets routed properly in a Microservices architecture and also that it waits until its picked up.
ECS with Autoscaling, will scale based on the unknown pattern of usage as mentioned.
upvoted 2 times

anibinaadi 1 year, 6 months ago

It is D
Refer https://aws.amazon.com/blogs/containers/amazon-elastic-container-service-ecs-auto-scaling-using-custom-metrics/ for additional information/knowledge.
upvoted 2 times

nosense 1 year, 6 months ago

Selected Answer: D

because it is scalable, reliable, and efficient.
C does not scale the models automatically
upvoted 4 times

deechean 1 year, 3 months ago

why C doesn't scale the model? Application Auto Scaling can apply to lambda.
upvoted 1 times

pentium75 11 months, 1 week ago

How would you "use Auto Scaling (!) to increase the number of vCPUs (!) for the Lamba functions"?
upvoted 3 times

NSA_Poker 6 months, 1 week ago

Auto Scaling doesn't apply to Lambda. As your functions receive more requests, Lambda automatically handles scaling the number of execution environments until you reach your account's concurrency limit.
upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #423 *Topic 1*

A solutions architect wants to use the following JSON text as an identity-based policy to grant specific permissions:

```
{    "Statement": [
        {
            "Action": [
                "ssm:ListDocuments",
                "ssm:GetDocument"
            ],
            "Effect": "Allow",
            "Resource": "*",
            "Sid": ""
        }
    ],
    "Version": "2012-10-17"
}
```

Which IAM principals can the solutions architect attach this policy to? (Choose two.)

A. Role **Most Voted**

B. Group **Most Voted**

C. Organization

D. Amazon Elastic Container Service (Amazon ECS) resource

E. Amazon EC2 resource

Correct Answer: *AB*

Community vote distribution

AB (100%)

Comments

nosense Highly Voted 1 year, 6 months ago

Selected Answer: AB

identity-based policy used for role and group

upvoted 19 times

pentium75 Highly Voted 11 months, 1 week ago

Selected Answer: AB

Isn't the content of the policy completely irrelevant? IAM policies are applied to users, groups or roles ...

upvoted 7 times

dkw2342 Most Recent 9 months ago

AB is correct, but the question is misleading because, according to the AWS IAM documentation, groups are not considered principals:
https://docs.aws.amazon.com/IAM/latest/UserGuide/intro-structure.html#intro-structure-principal."

upvoted 2 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: AB

A. Role
B. Group

upvoted 3 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: AB

Role or group

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #424 *Topic 1*

A company is running a custom application on Amazon EC2 On-Demand Instances. The application has frontend nodes that need to run 24 hours a day, 7 days a week and backend nodes that need to run only for a short time based on workload. The number of backend nodes varies during the day.

The company needs to scale out and scale in more instances based on workload.

Which solution will meet these requirements MOST cost-effectively?

A. Use Reserved Instances for the frontend nodes. Use AWS Fargate for the backend nodes.

B. Use Reserved Instances for the frontend nodes. Use Spot Instances for the backend nodes. Most Voted

C. Use Spot Instances for the frontend nodes. Use Reserved Instances for the backend nodes.

D. Use Spot Instances for the frontend nodes. Use AWS Fargate for the backend nodes.

Correct Answer: *B*

Community vote distribution

B (62%) A (38%)

Comments

Ramdi1 Highly Voted 1 year, 8 months ago

Selected Answer: A

Has to be A, It can scale down if required and you will be charged for what you use with fargate. Secondly they have not said the backend can have timeouts or can be down for a little period of time or something. So it has to rule out any spot instances even though they are cheaper.

upvoted 18 times

awsgeek75 1 year, 5 months ago

Fargate is serverless EKS so it cannot manage EC2 nodes

upvoted 7 times

nosense Highly Voted 2 years ago

Selected Answer: B

Reserved + spot

Reserved+ spot .
Fargate for serverless
upvoted 15 times

Andreew883 Most Recent 1 month, 3 weeks ago

Selected Answer: B

The questione Is about instances so container and fairgate Is not applicabile and we Need to choose respojse B
upvoted 1 times

zdi561 4 months ago

Selected Answer: A

I think in A it should be Fargat spot, B means standard spot. The former uses auto scaling and the later does not
upvoted 1 times

zdi561 4 months, 2 weeks ago

Selected Answer: A

The requirement is to have an autoscaling , and fargate can do that. Spot instance only works when the work load is fault tolerant. Furthermore it is told to use on-demand instance.
upvoted 2 times

Mayur_B 7 months, 4 weeks ago

I will go with A.
Where does the question mentions that the workload interruptions are accepted.
upvoted 3 times

mussha 1 year, 3 months ago

Selected Answer: B

B) because firegate is containser
upvoted 3 times

noircesar25 1 year, 3 months ago

so what ive make up from this scenario is: the key word right here is "backend nodes" you cant use a serverless compute service with nodes and you need to use EC2s
so if we had ECS EC2 lunch type or on-demand EC2s as an options for the backend, they would be true?
upvoted 2 times

mwwt2022 1 year, 5 months ago

Selected Answer: B

24-7 usage for fe -> reserved instance
irregular workload for be -> spot instance
upvoted 4 times

pentium75 1 year, 5 months ago

Selected Answer: B

Not A because Fargate runs containers, not EC2 instances. But we have no indication that the workload would be containerized; it runs "on EC2 instances".
Not C and D because frontend must run 24/7, can't use Spot.

Thus B, yes, Spot instances are risky, but as they need to run "only for a short time" it seems acceptable.

Technically ideal option would be Reserved Instances for frontend nodes and On-demand instances for backend nodes, but that is not an option here.
upvoted 8 times

[Removed] 1 year, 6 months ago

Selected Answer: B

Not sure the application can be containerized
upvoted 3 times

AwsZora 1 year, 6 months ago

Selected Answer: A

it is safe

upvoted 1 times

awsgeek75 1 year, 4 months ago

Fargate = containers
A is wrong

upvoted 2 times

meowruki 1 year, 6 months ago

Selected Answer: B

Reserved Instances (RIs) for Frontend Nodes: Since the frontend nodes need to run continuously (24/7), using Reserved Instances for them makes sense. RIs provide significant cost savings compared to On-Demand Instances for steady-state workloads.

Spot Instances for Backend Nodes: Spot Instances are suitable for short-duration workloads and can be significantly cheaper than On-Demand Instances. Since the number of backend nodes varies during the day, Spot Instances can help you take advantage of spare capacity at a lower cost. Keep in mind that Spot Instances may be interrupted if the capacity is needed elsewhere, so they are best suited for stateless and fault-tolerant workloads.

upvoted 2 times

meowruki 1 year, 6 months ago

Option A (Use Reserved Instances for the frontend nodes. Use AWS Fargate for the backend nodes): AWS Fargate is a serverless compute engine for containers, and it may not be the best fit for the described backend workload, especially if the number of backend nodes varies during the day.

upvoted 1 times

Goutham4981 1 year, 6 months ago

Selected Answer: B

AWS Fargate is a serverless compute engine for containers that allows you to run containers without having to manage the underlying infrastructure. It simplifies the process of deploying and managing containerized applications by abstracting away the complexities of server management, scaling, and cluster orchestration.
No containerized application requirements are mentioned in the question. Plain EC2 instances. So Fargate is not actually an option

upvoted 3 times

thanhnv142 1 year, 7 months ago

A is fargate, which is none sense. B seems more OK (though none-sense)

upvoted 4 times

dilaaziz 1 year, 8 months ago

Selected Answer: A

Fargate for backend node

upvoted 2 times

awsgeek75 1 year, 4 months ago

Fargate is for containers not EC2 so A is wrong

upvoted 2 times

Wayne23Fang 1 year, 8 months ago

Selected Answer: A

(B) would take chance, though unlikely (A) is server-less auto-scaling. In case backend is idle, it might scale down, save money but no need to worry for interruption by Spot instance.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #425 *Topic 1*

A company uses high block storage capacity to runs its workloads on premises. The company's daily peak input and output transactions per second are not more than 15,000 IOPS. The company wants to migrate the workloads to Amazon EC2 and to provision disk performance independent of storage capacity.

Which Amazon Elastic Block Store (Amazon EBS) volume type will meet these requirements MOST cost-effectively?

A. GP2 volume type

B. io2 volume type

C. GP3 volume type **Most Voted**

D. io1 volume type

Correct Answer: *C*

Community vote distribution

C (96%) B

Comments

nosense Highly Voted 2 years ago

Selected Answer: C

Gp3 $ 0.08 usd per gb
Gp2 $ 0.10 usd per gb

upvoted 15 times

Yadav_Sanjay Highly Voted 2 years ago

Selected Answer: C

Both GP2 and GP3 has max IOPS 16000 but GP3 is cost effective.
https://aws.amazon.com/blogs/storage/migrate-your-amazon-ebs-volumes-from-gp2-to-gp3-and-save-up-to-20-on-costs/

upvoted 12 times

LeonSauveterre Most Recent 6 months, 1 week ago

Selected Answer: C

OK, and is this something I should know to prepare for the exam...

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: C

C. GP3 volume type

upvoted 4 times

james2033 1 year, 10 months ago

Selected Answer: C

Quote "customers can scale up to 16,000 IOPS and" at https://aws.amazon.com/about-aws/whats-new/2020/12/introducing-new-amazon-ebs-general-purpose-volumes-gp3/

upvoted 3 times

alexandercamachop 2 years ago

Selected Answer: C

The GP3 (General Purpose SSD) volume type in Amazon Elastic Block Store (EBS) is the most cost-effective option for the given requirements. GP3 volumes offer a balance of price and performance and are suitable for a wide range of workloads, including those with moderate I/O needs.

GP3 volumes allow you to provision performance independently from storage capacity, which means you can adjust the baseline performance (measured in IOPS) and throughput (measured in MiB/s) separately from the volume size. This flexibility allows you to optimize your costs while meeting the workload requirements.

In this case, since the company's daily peak input and output transactions per second are not more than 15,000 IOPS, GP3 volumes provide a suitable and cost-effective option for their workloads.

upvoted 2 times

maver144 2 years ago

Selected Answer: B

It is not C pals. The company wants to migrate the workloads to Amazon EC2 and to provision disk performance independent of storage capacity. With GP3 we have to increase storage capacity to increase IOPS over baseline.

You can only chose IOPS independetly with IO family and IO2 is in general better then IO1.

upvoted 2 times

somsundar 1 year, 10 months ago

@maver144 - That's the case with GP2 volumes. With GP3 we can define IOPS independent of storage capacity.

upvoted 4 times

Joselucho38 2 years ago

Selected Answer: C

Therefore, the most suitable and cost-effective option in this scenario is the GP3 volume type (option C).

upvoted 2 times

Efren 2 years ago

Selected Answer: C

GPS3 allows 16000 IOPS

upvoted 4 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #426 *Topic 1*

A company needs to store data from its healthcare application. The application's data frequently changes. A new regulation requires audit access at all levels of the stored data.

The company hosts the application on an on-premises infrastructure that is running out of storage capacity. A solutions architect must securely migrate the existing data to AWS while satisfying the new regulation.

Which solution will meet these requirements?

A. Use AWS DataSync to move the existing data to Amazon S3. Use AWS CloudTrail to log data events. **Most Voted**

B. Use AWS Snowcone to move the existing data to Amazon S3. Use AWS CloudTrail to log management events.

C. Use Amazon S3 Transfer Acceleration to move the existing data to Amazon S3. Use AWS CloudTrail to log data events.

D. Use AWS Storage Gateway to move the existing data to Amazon S3. Use AWS CloudTrail to log management events.

Correct Answer: *A*

Community vote distribution

A (69%) D (31%)

Comments

thanhnv142 Highly Voted 1 year, 7 months ago

A is better because:
- Data sync is used for migrate. Storage gw is used to connect on-prem to AWS.
- dataevents is to log for access, management events is for config or management

upvoted 14 times

pentium75 Highly Voted 1 year, 5 months ago

Selected Answer: A

We need to log "access at all levels" aka "data events", thus B and D are out (logging only "management events" like granting permissions or changing the access tier).
C, S3 Transfer Acceleration is to increase upload performance from widespread sources or over unreliable networks, but it just provides an endpoint, it does not upload anything itself.

upvoted 11 times

hpirnaj Most Recent 5 months ago

Selected Answer: A

data management shows management operations that are performed on resources in your AWS account so, D is wrong
https://repost.aws/knowledge-center/cloudtrail-data-management-events

upvoted 2 times

JA2018 6 months, 2 weeks ago

Selected Answer: A

Keys in STEM:

A company needs to store data from its healthcare application. The application's data frequently changes. A new regulation requires audit access at all levels of the stored data.

1. Must securely migrate the existing data to AWS while satisfying the new regulation.

2. Logs access at all data levels

upvoted 2 times

1e22522 10 months, 1 week ago

Selected Answer: A

Typical DataSync scenario me thinks!

upvoted 3 times

osmk 1 year, 3 months ago

Selected Answer: A

Use AWS DataSync to migrate existing data to Amazon S3https://aws.amazon.com/datasync/faqs/

upvoted 4 times

NayeraB 1 year, 3 months ago

Selected Answer: A

It's DataSync for me

upvoted 3 times

frmrkc 1 year, 4 months ago

Selected Answer: D

Storage Gateway integration with CloudTrail :
https://docs.aws.amazon.com/filegateway/latest/filefsxw/logging-using-cloudtrail.html

whereas DataSync can be monitored with Amazon CloudWatch:
https://docs.aws.amazon.com/datasync/latest/userguide/monitor-datasync.html

upvoted 2 times

frmrkc 1 year, 4 months ago

and here are all Storage Gateway actions monitored by CloudTrail:
https://docs.aws.amazon.com/storagegateway/latest/APIReference/API_Operations.html

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: A

B and C don't solve the problem
A is extending the data and management events are for administrative actions only (tracking account creation, user security actions etc.).
C uses DataSync to move all the data and logs data events which include S3 file uploads and downloads.

Management events: User logs into an EC2 instance, creates an S3 IAM role
Data events: User uploads a file to S3

upvoted 4 times

benacert 1 year, 5 months ago

A- DataSync secure fast data transfer

upvoted 2 times

upvoted 2 times

ZZZ_Sleep 1 year, 5 months ago

Selected Answer: D

Keyword of this question = running out of storage capacity

AWS Storage Gateway = extend the on-premises storage
AWS DataSync = copy data between on-premises storage

So, the answer should be D (AWS Storage Gateway)

upvoted 6 times

pentium75 1 year, 5 months ago

"Securely migrate the existing data to AWS" -> move data away fron on-premises storage to AWS. Plus, D logs only management events, not "access at all levels".

upvoted 6 times

aws94 1 year, 5 months ago

Selected Answer: D

AWS DataSync is designed for fast, simple, and secure data transfer, but it focuses more on data synchronization rather than on-premises migration.

upvoted 1 times

pentium75 1 year, 5 months ago

Thus it is wrong, but more because of the incorrect logging option in this answer.

upvoted 1 times

meowruki 1 year, 6 months ago

Selected Answer: A

AWS DataSync is suitable for data transfer and synchronization

Option D (Use AWS Storage Gateway to move the existing data to Amazon S3. Use AWS CloudTrail to log management events): AWS Storage Gateway is typically used for hybrid cloud storage solutions and may introduce additional complexity for a one-time data migration task. It might not be as straightforward as using AWS Snowcone for this specific scenario.

upvoted 2 times

chikuwan 1 year, 6 months ago

Selected Answer: A

both DataSync and Storage Gateway are fine to sync data...but to "audit access at all levels of the stored data" ...it should be data events(data plane operation)..management event is some account level things.
So answer should be A

upvoted 3 times

bogobob 1 year, 6 months ago

Selected Answer: D

While both DataSync and Storage Gateway allow syncing of data between on-premise and cloud, DataSync is built for rapid shifting of data into a cloud environment, not specifically for continued use in on-premise servers.

upvoted 2 times

potomac 1 year, 7 months ago

Selected Answer: A

AWS DataSync is an online data transfer service that simplifies, automates, and accelerates the process of copying large amounts of data to and from AWS storage services over the Internet or over AWS Direct Connect.

upvoted 2 times

pentium75 1 year, 5 months ago

What about logging?

upvoted 2 times

JA2018 6 months, 2 weeks ago

This is where AWS CloudTrail comes into play.
upvoted 1 times

canonlycontainletters1 1 year, 7 months ago

Selected Answer: A

A seems to be more convincing to me.
upvoted 2 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #427 *Topic 1*

A solutions architect is implementing a complex Java application with a MySQL database. The Java application must be deployed on Apache Tomcat and must be highly available.

What should the solutions architect do to meet these requirements?

A. Deploy the application in AWS Lambda. Configure an Amazon API Gateway API to connect with the Lambda functions.

B. Deploy the application by using AWS Elastic Beanstalk. Configure a load-balanced environment and a rolling deployment policy. **Most Voted**

C. Migrate the database to Amazon ElastiCache. Configure the ElastiCache security group to allow access from the application.

D. Launch an Amazon EC2 instance. Install a MySQL server on the EC2 instance. Configure the application on the server. Create an AMI. Use the AMI to create a launch template with an Auto Scaling group.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

cloudenthusiast Highly Voted 2 years ago

B
AWS Elastic Beanstalk provides an easy and quick way to deploy, manage, and scale applications. It supports a variety of platforms, including Java and Apache Tomcat. By using Elastic Beanstalk, the solutions architect can upload the Java application and configure the environment to run Apache Tomcat.

upvoted 11 times

KennethNg923 Most Recent 11 months, 4 weeks ago

Selected Answer: B

Beanstalk for sure

upvoted 2 times

zinabu 1 year, 1 month ago

By using Elastic Beanstalk, the solutions architect can upload the Java application and configure the environment to run Apache Tomcat

Tomcat.
upvoted 3 times

wizcloudifa 1 year, 1 month ago

Selected Answer: B

The key word here from the question if you notice is "The Java application must be DEPLOYED..." hence Elastic Beanstalk, it is a serverless deployment service and supports a variety of platforms(apache Tomcat in our situation), and it will scale automatically with less operational overhead(unlike option D with a lot of operation overhead)
upvoted 3 times

awsgeek75 1 year, 4 months ago

Selected Answer: B

https://aws.amazon.com/elasticbeanstalk/details/
upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

B. Deploy the application by using AWS Elastic Beanstalk. Configure a load-balanced environment and a rolling deployment policy.
upvoted 4 times

james2033 1 year, 10 months ago

Selected Answer: B

Keyword "AWS Elastic Beanstalk" for re-architecture from Java web-app inside Apache Tomcat to AWS Cloud.
upvoted 3 times

TariqKipkemei 2 years ago

Selected Answer: B

Definitely B
upvoted 2 times

antropaws 2 years ago

Selected Answer: B

Clearly B.
upvoted 2 times

nosense 2 years ago

Selected Answer: B

Easy deploy, management and scale
upvoted 3 times

greyrose 2 years ago

Selected Answer: B

BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #428 *Topic 1*

A serverless application uses Amazon API Gateway, AWS Lambda, and Amazon DynamoDB. The Lambda function needs permissions to read and write to the DynamoDB table.

Which solution will give the Lambda function access to the DynamoDB table MOST securely?

A. Create an IAM user with programmatic access to the Lambda function. Attach a policy to the user that allows read and write access to the DynamoDB table. Store the access_key_id and secret_access_key parameters as part of the Lambda environment variables. Ensure that other AWS users do not have read and write access to the Lambda function configuration.

B. Create an IAM role that includes Lambda as a trusted service. Attach a policy to the role that allows read and write access to the DynamoDB table. Update the configuration of the Lambda function to use the new role as the execution role. **Most Voted**

C. Create an IAM user with programmatic access to the Lambda function. Attach a policy to the user that allows read and write access to the DynamoDB table. Store the access_key_id and secret_access_key parameters in AWS Systems Manager Parameter Store as secure string parameters. Update the Lambda function code to retrieve the secure string parameters before connecting to the DynamoDB table.

D. Create an IAM role that includes DynamoDB as a trusted service. Attach a policy to the role that allows read and write access from the Lambda function. Update the code of the Lambda function to attach to the new role as an execution role.

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

awsgeek75 Highly Voted 1 year, 5 months ago

Selected Answer: B

DynamoDB needs to trust Lambda. NOT the other way around. So Lambda must be configured as a trusted service. Role for service which gives B and D options. D is setting up (somehow?) to allow Lambda to trust DynamoDB... or the wording makes no sense.

upvoted 8 times

james2033 Highly Voted 1 year, 10 months ago

Selected Answer: B

Keyword B. " IAM role that includes Lambda as a trusted service", not "IAM role that includes DynamoDB as a trusted service" in D. It is IAM role, not IAM user.

upvoted 6 times

KennethNg923 Most Recent 11 months, 4 weeks ago

Selected Answer: B

IAM Role for access to DynamoDB, not for access Lambda

upvoted 2 times

antropaws 2 years ago

Selected Answer: B

B sounds better.

upvoted 3 times

omoakin 2 years ago

BBBBBBBBBB

upvoted 2 times

alvinnguyennexcel 2 years ago

Selected Answer: B

vote B

upvoted 2 times

cloudenthusiast 2 years ago

B
Option B suggests creating an IAM role that includes Lambda as a trusted service, meaning the role is specifically designed for Lambda functions. The role should have a policy attached to it that grants the required read and write access to the DynamoDB table.

upvoted 4 times

nosense 2 years ago

Selected Answer: B

B is right
Role key word and trusted service lambda

upvoted 5 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #429 *Topic 1*

The following IAM policy is attached to an IAM group. This is the only policy applied to the group.

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "1",
            "Effect": "Allow",
            "Action": "ec2:*",
            "Resource": "*",
            "Condition": {
                "StringEquals": {
                    "ec2:Region": "us-east-1"
                }
            }
        },
        {
            "Sid": "2",
            "Effect": "Deny",
            "Action": [
                "ec2:StopInstances",
                "ec2:TerminateInstances"
            ],
            "Resource": "*",
            "Condition": {
                "BoolIfExists": {"aws:MultiFactorAuthPresent": false}
            }
        }
    ]
}
```

What are the effective IAM permissions of this policy for group members?

A. Group members are permitted any Amazon EC2 action within the us-east-1 Region. Statements after the Allow permission are not applied.

B. Group members are denied any Amazon EC2 permissions in the us-east-1 Region unless they are logged in with multi-factor authentication (MFA).

C. Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for all Regions when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action.

D. Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for the us-east-1 Region only when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action within the us-east-1 Region. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (90%) C (10%)

Comments

jack79 Highly Voted 1 year, 11 months ago

came in exam today

upvoted 13 times

KennethNg923 Most Recent 11 months, 3 weeks ago

Selected Answer: D

for the us-east-1 Region only, not for all region

upvoted 2 times

wizcloudifa 1 year, 1 month ago

Selected Answer: D

One of the few situations when actual answer is same as the most voted answer lol

upvoted 2 times

pdragon1981 1 year, 5 months ago

Selected Answer: C

Not sure why everyone vote D, I think that the valid option as to be C as the second condition regarding MFA there is point that only refer to a specific region, so basically this means that is for all the regions

upvoted 2 times

pdragon1981 1 year, 5 months ago

Ok ignore D is right as the first condition is what gives permission to make anything for EC2 but is restricted to us-east-1 region

upvoted 6 times

youdelin 1 year, 7 months ago

the json is describing a lot of things apparently, so I go with the longest answer lol

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: D

D. Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for the us-east-1 Region only when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action within the us-east-1 Region

upvoted 3 times

james2033 1 year, 10 months ago

Selected Answer: D

A. "Statements after the Allow permission are not applied." --> Wrong.

B. "denied any Amazon EC2 permissions in the us-east-1 Region" --> Wrong. Just deny 2 items.

C. "allowed the ec2:StopInstances and ec2:TerminateInstances permissions for all Regions" --> Wrong. Just region us-east-1.

D. ok.

upvoted 2 times

TariqKipkemei 2 years ago

Selected Answer: D

Only D makes sense

upvoted 2 times

antropaws 2 years ago

Selected Answer: D

D sounds about right.

upvoted 2 times

alvinnguyennexcel 2 years ago

Selected Answer: D

D is correct

upvoted 3 times

omoakin 2 years ago

D is correct

upvoted 2 times

nosense 2 years ago

Selected Answer: D

D is right

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #430 *Topic 1*

A manufacturing company has machine sensors that upload .csv files to an Amazon S3 bucket. These .csv files must be converted into images and must be made available as soon as possible for the automatic generation of graphical reports.

The images become irrelevant after 1 month, but the .csv files must be kept to train machine learning (ML) models twice a year. The ML trainings and audits are planned weeks in advance.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

A. Launch an Amazon EC2 Spot Instance that downloads the .csv files every hour, generates the image files, and uploads the images to the S3 bucket.

B. Design an AWS Lambda function that converts the .csv files into images and stores the images in the S3 bucket. Invoke the Lambda function when a .csv file is uploaded. **Most Voted**

C. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 Glacier 1 day after they are uploaded. Expire the image files after 30 days. **Most Voted**

D. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) 1 day after they are uploaded. Expire the image files after 30 days.

E. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) 1 day after they are uploaded. Keep the image files in Reduced Redundancy Storage (RRS).

Correct Answer: *BC*

Community vote distribution

BC (90%) | BE (10%)

Comments

vesen22 Highly Voted 1 year, 6 months ago

Selected Answer: BC

https://docs.aws.amazon.com/amazonglacier/latest/dev/introduction.html

upvoted 5 times

awsgeek75 Most Recent 11 months, 1 week ago

Selected Answer: BC

B for processing the images via Lambda as it's more cost efficient than EC2 spot instances
C for expiring images after 30 days and because the ML trainings are planned weeks in advance so S3 glacier is ideal for slow retrieval and cheap storage.

D and E uses S3 infrequent access which is more expensive than glacier

upvoted 3 times

pentium75 11 months, 1 week ago

Selected Answer: BC

Not A, we need the images "as soon as possible", A runs every hour
"ML trainings and audits are planned weeks in advance" thus Glacier (C) is ok.

upvoted 3 times

Xin123 1 year, 2 months ago

Selected Answer: BC

Answer is B&C. For D, you must store data for 30 days in s3 standard before move to IA tiers, glacier is fine

https://docs.aws.amazon.com/AmazonS3/latest/userguide/lifecycle-transition-general-considerations.html#:~:text=Before%20you%20transition%20objects%20to%20S3%20Standard%2DIA%20or%20S3%20One%20Zone%2DIA%2C%20you%20must%20store%20them%20for%20at%20least%2030%20days%20in%20Amazon%20S3

upvoted 4 times

Guru4Cloud 1 year, 3 months ago

Selected Answer: BC

Definitely B & C

upvoted 3 times

jayce5 1 year, 4 months ago

Selected Answer: BC

A. Wrong, the .csv files must be processed asap.
D and E are incorrect since Glacier is the most cost-effective option, and plans for using .csv files are known weeks in advance.

upvoted 2 times

james2033 1 year, 4 months ago

Why need "These .csv files must be converted into images"?

upvoted 1 times

awsgeek75 10 months, 3 weeks ago

Because they are being used in some graphical reports (probably fancy powerpoint presentations!)

upvoted 2 times

smartegnine 1 year, 5 months ago

Selected Answer: BC

the key word is Weeks in advance, even you save data in S3 Gracia will also OK to take couples days to retrieve the data

upvoted 3 times

TariqKipkemei 1 year, 6 months ago

Selected Answer: BC

Definitely B & C

upvoted 2 times

Abrar2022 1 year, 6 months ago

Selected Answer: BC

A. Wrong because Lifecycle rule is not mentioned.

B. CORRECT

C. CORRECT

D. Why Store on S3 One Zone-Infrequent Access (S3 One Zone-IA) when the files are going to irrelevant after 1 month? (Availability 99.99% - consider cost)

E. again, Why use Reduced Redundancy Storage (RRS) when the files are irrelevant after 1 month? (Availability 99.99% - consider cost)

upvoted 4 times

RoroJ 1 year, 6 months ago

Selected Answer: BE

B: Serverless and fast responding
E: will keep .csv file for a year, C and D expires the file after 30 days.

upvoted 4 times

RoroJ 1 year, 6 months ago

B&C, misread the question, expires the image files after 30 days.

upvoted 3 times

hiroohiroo 1 year, 6 months ago

Selected Answer: BC

https://aws.amazon.com/jp/about-aws/whats-new/2021/11/amazon-s3-glacier-storage-class-amazon-s3-glacier-flexible-retrieval/

upvoted 3 times

nosense 1 year, 6 months ago

Selected Answer: BC

B severless and cost effective
C corrctl rule to store

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #431 *Topic 1*

A company has developed a new video game as a web application. The application is in a three-tier architecture in a VPC with Amazon RDS for MySQL in the database layer. Several players will compete concurrently online. The game's developers want to display a top-10 scoreboard in near-real time and offer the ability to stop and restore the game while preserving the current scores.

What should a solutions architect do to meet these requirements?

A. Set up an Amazon ElastiCache for Memcached cluster to cache the scores for the web application to display.

B. Set up an Amazon ElastiCache for Redis cluster to compute and cache the scores for the web application to display. **Most Voted**

C. Place an Amazon CloudFront distribution in front of the web application to cache the scoreboard in a section of the application.

D. Create a read replica on Amazon RDS for MySQL to run queries to compute the scoreboard and serve the read traffic to the web application.

Correct Answer: *B*

Community vote distribution

B (92%) A (8%)

Comments

Guru4Cloud Highly Voted 1 year, 3 months ago

Selected Answer: B

Redis provides fast in-memory data storage and processing. It can compute the top 10 scores and update the cache in milliseconds.
ElastiCache Redis supports sorting and ranking operations needed for the top 10 leaderboard.
The cached leaderboard can be retrieved from Redis vs hitting the MySQL database for every read. This reduces load on the database.
Redis supports persistence, so scores are preserved if the cache stops/restarts

upvoted 10 times

TariqKipkemei Highly Voted 1 year, 1 month ago

Selected Answer: B

Real-time gaming leaderboards are easy to create with Amazon ElastiCache for Redis. Just use the Redis Sorted Set data structure, which provides uniqueness of elements while maintaining the list sorted by their scores. Creating a real-time ranked list is as simple as updating a user's score each time it changes. You can also use Sorted Sets to handle time series data by using timestamps as the score.

https://aws.amazon.com/elasticache/redis/#:~:text=ElastiCache%20for%20Redis.-,Gaming,-Leaderboards

upvoted 8 times

potomac Most Recent 1 year, 1 month ago

Selected Answer: B

ElastiCache for Redis sorts and ranks datasets

upvoted 5 times

5ab5e39 1 year, 3 months ago

https://aws.amazon.com/blogs/database/building-a-real-time-gaming-leaderboard-with-amazon-elasticache-for-redis/

upvoted 4 times

ukivanlamlpi 1 year, 3 months ago

Selected Answer: A

concurrently = memcached

upvoted 3 times

awsgeek75 11 months ago

Amazon ElastiCache for Memcached is non-persistent so start/stop of game will lose the score." and offer the ability to stop and restore the game while preserving the current scores."

upvoted 3 times

james2033 1 year, 4 months ago

Selected Answer: B

See case study of leaderboard with Redis at https://redis.io/docs/data-types/sorted-sets/ , it is feature "sorted sets". See comparison between Redis an d Memcached at https://docs.aws.amazon.com/AmazonElastiCache/latest/mem-ug/SelectEngine.html , the different at feature "Sorted sets"

upvoted 4 times

live_reply_developers 1 year, 5 months ago

Selected Answer: B

advanced data structures, complex querying, pub/sub messaging, or persistence, Redis may be a better fit.

upvoted 2 times

haoAWS 1 year, 5 months ago

B is correct

upvoted 2 times

jf_topics 1 year, 5 months ago

B correct.

upvoted 2 times

hiroohiroo 1 year, 6 months ago

Selected Answer: B

https://aws.amazon.com/jp/blogs/news/building-a-real-time-gaming-leaderboard-with-amazon-elasticache-for-redis/

upvoted 4 times

cloudenthusiast 1 year, 6 months ago

Amazon ElastiCache for Redis is a highly scalable and fully managed in-memory data store. It can be used to store and compute the scores in real time for the top-10 scoreboard. Redis supports sorted sets, which can be used to store the scores as well as perform efficient queries to retrieve the top scores. By utilizing ElastiCache for Redis, the web application can quickly retrieve the current scores without the need to perform complex and potentially resource-intensive database queries.

upvoted 3 times

nosense 1 year, 6 months ago

Selected Answer: B

B is right

upvoted 2 times

Efren 1 year, 6 months ago

More questions!!!

upvoted 5 times